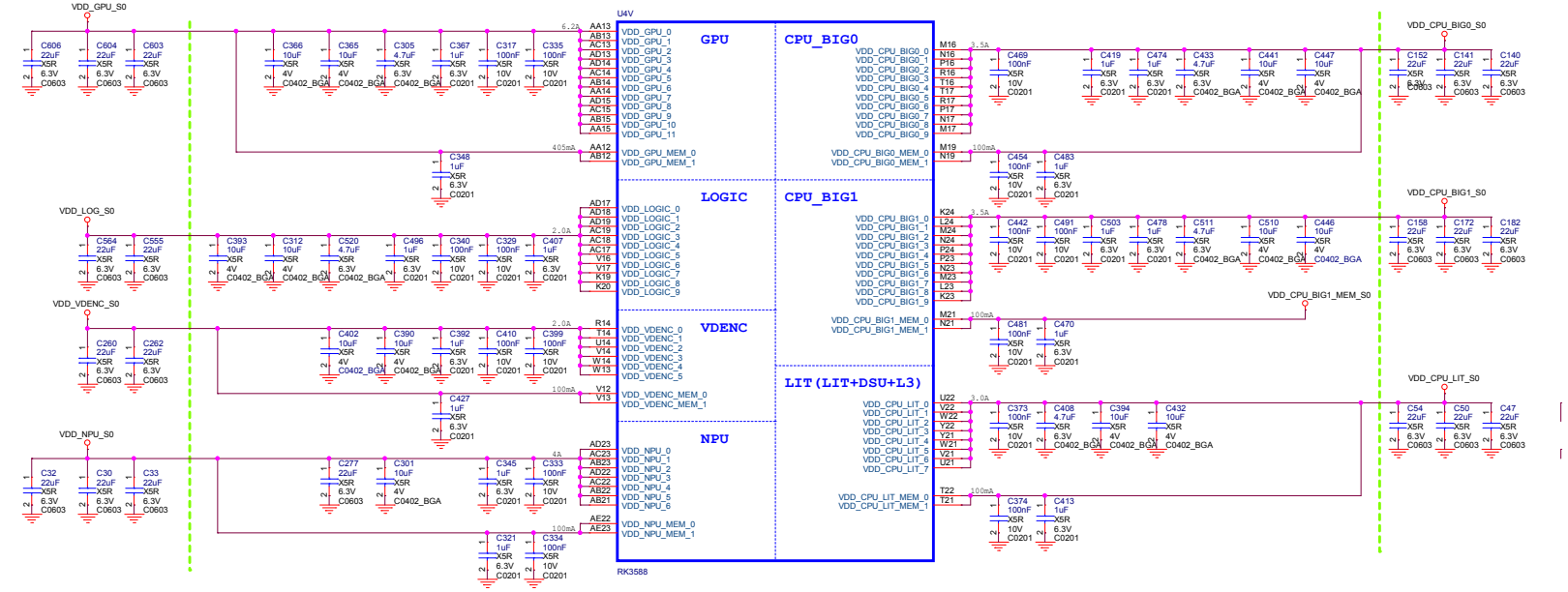
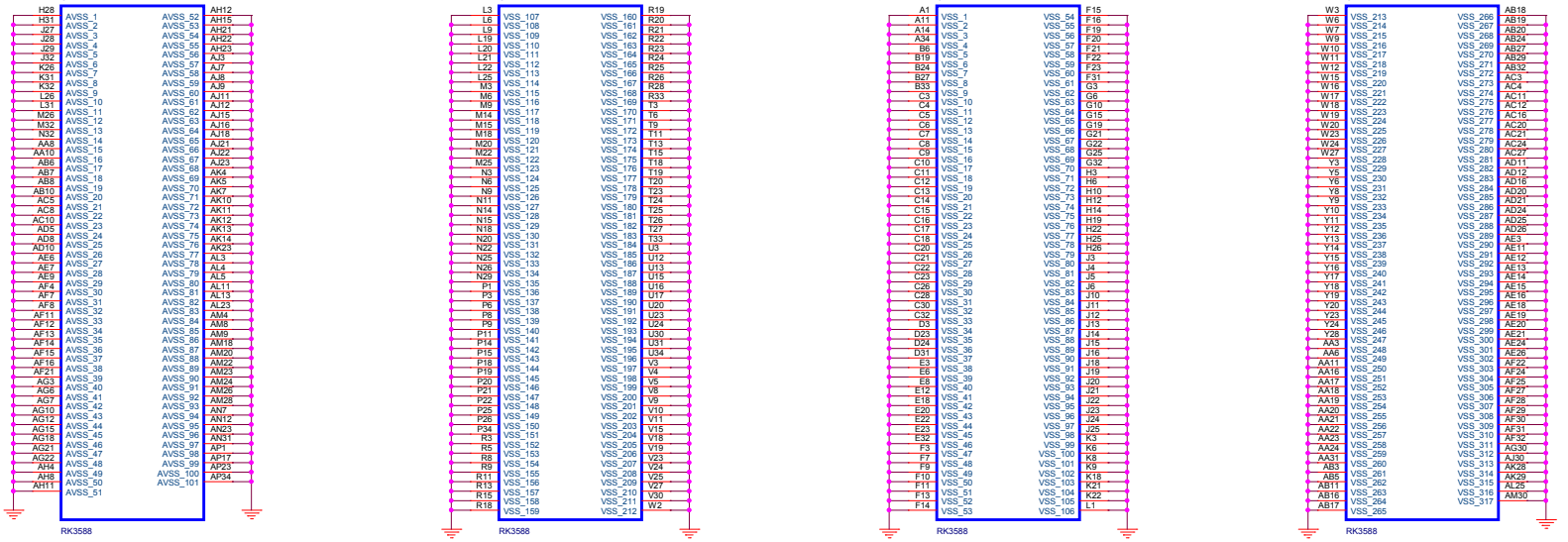
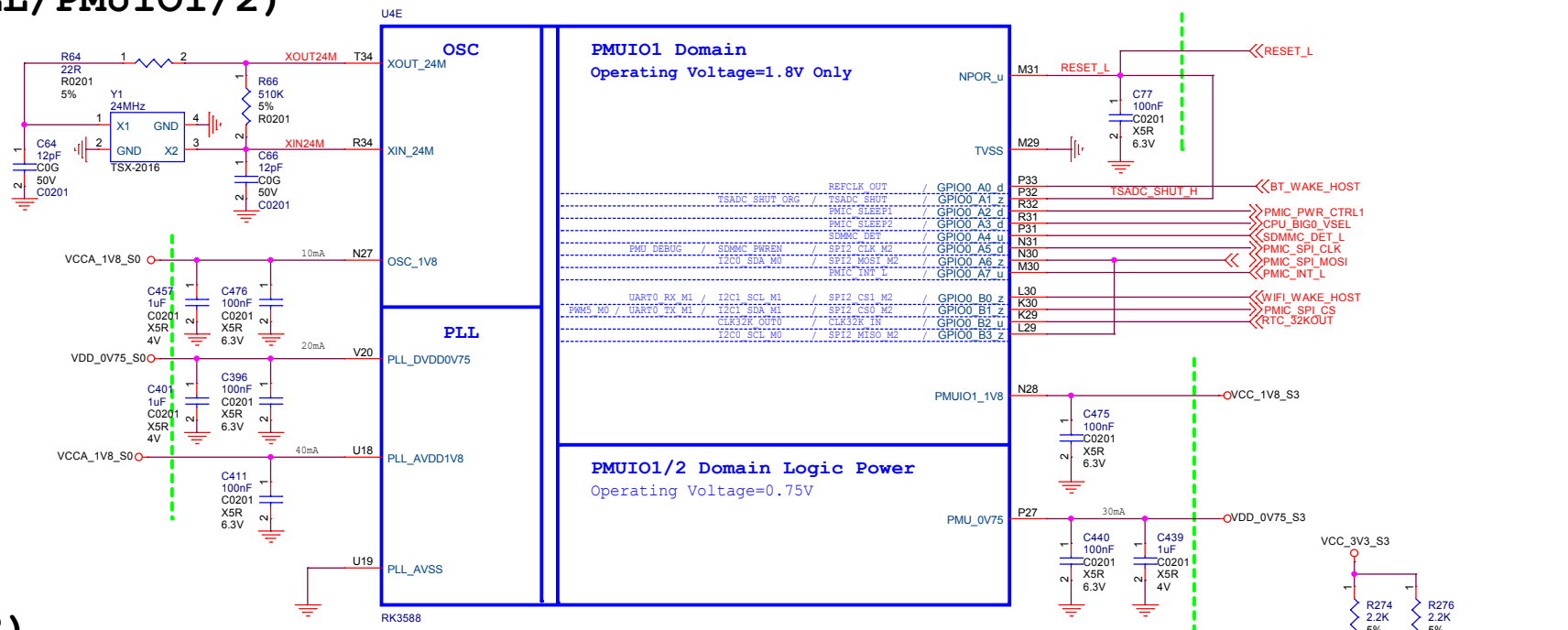
RK3588_V(POWER)



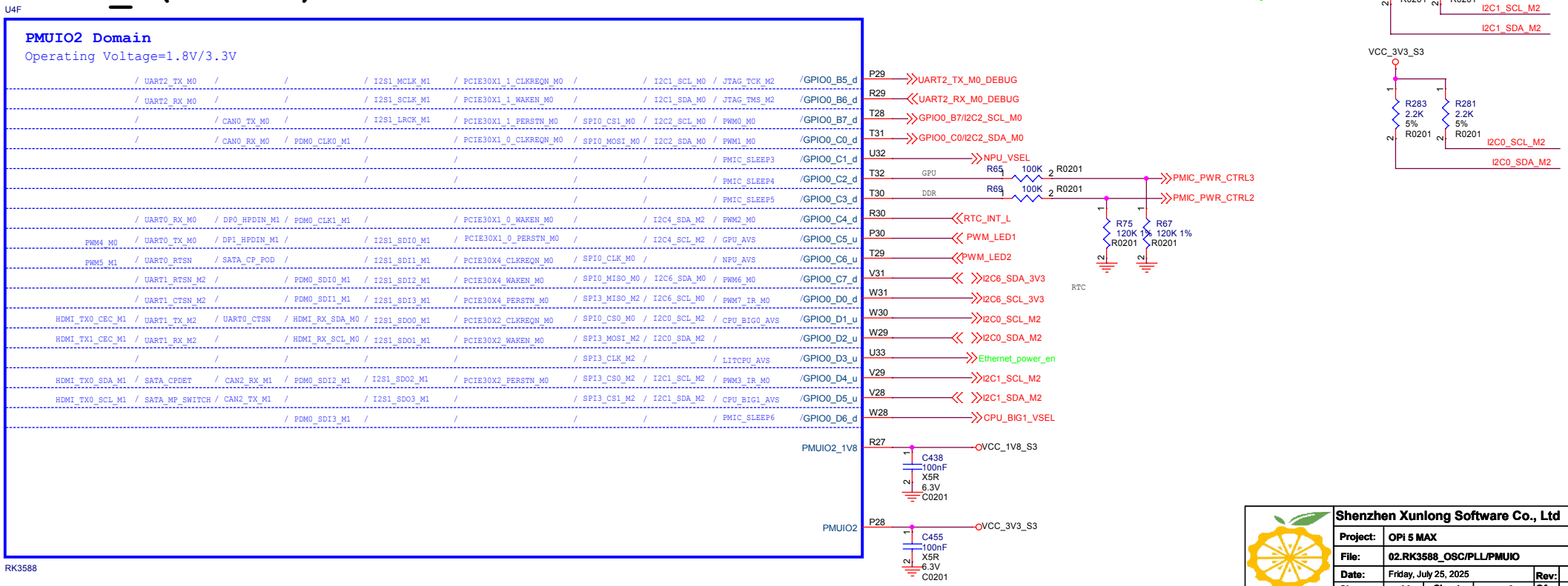


Shenzhen Xunlong Software Co., Ltd	
Project:	OPi 5 MAX
File:	01.RK3588_Power/GND
Date:	Friday, July 25, 2025
Rev:	V1.1
Size:	A3
Sheet:	1
Of:	29

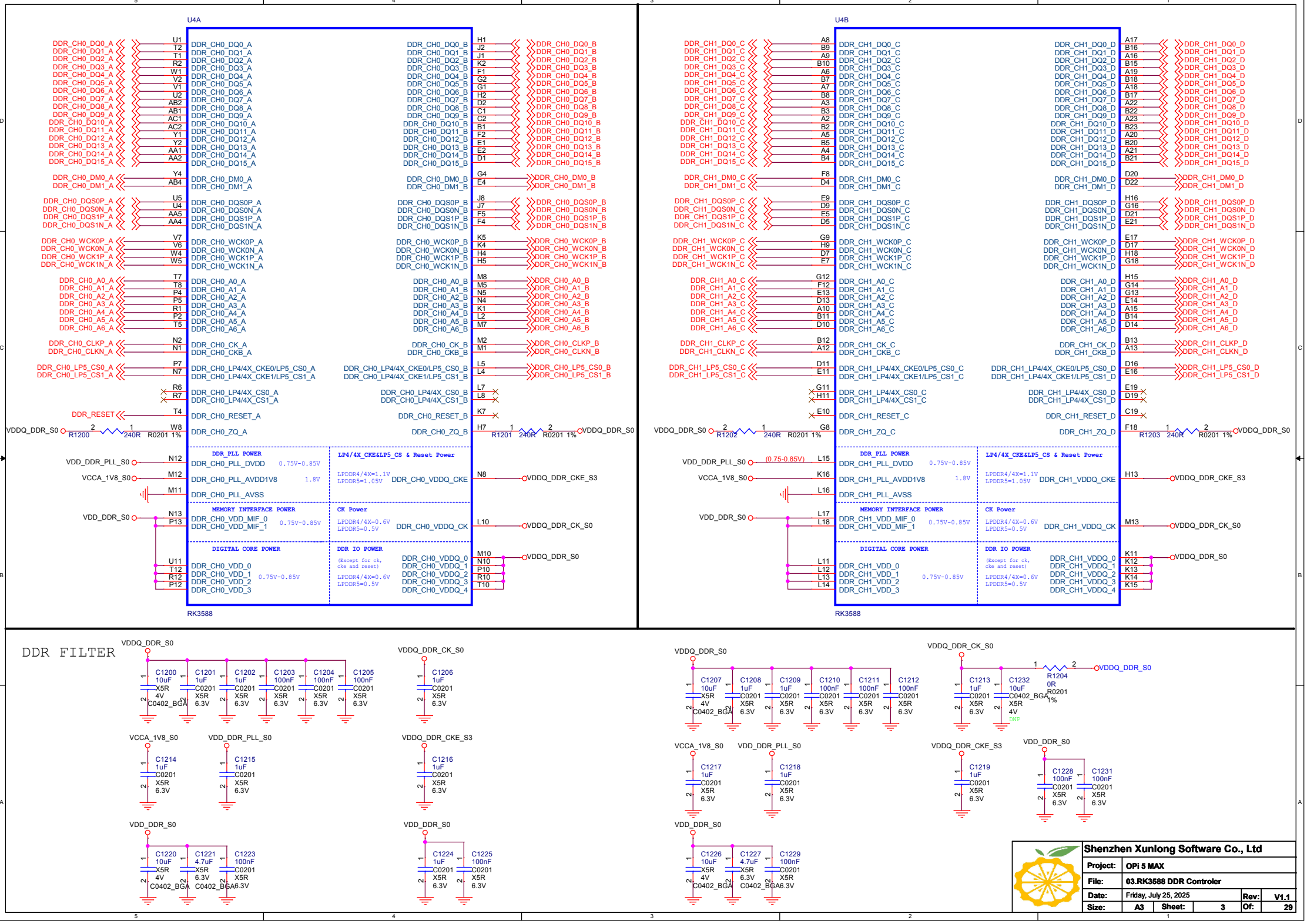
RK3588_E(OSC/PLL/PMUIO1/2)



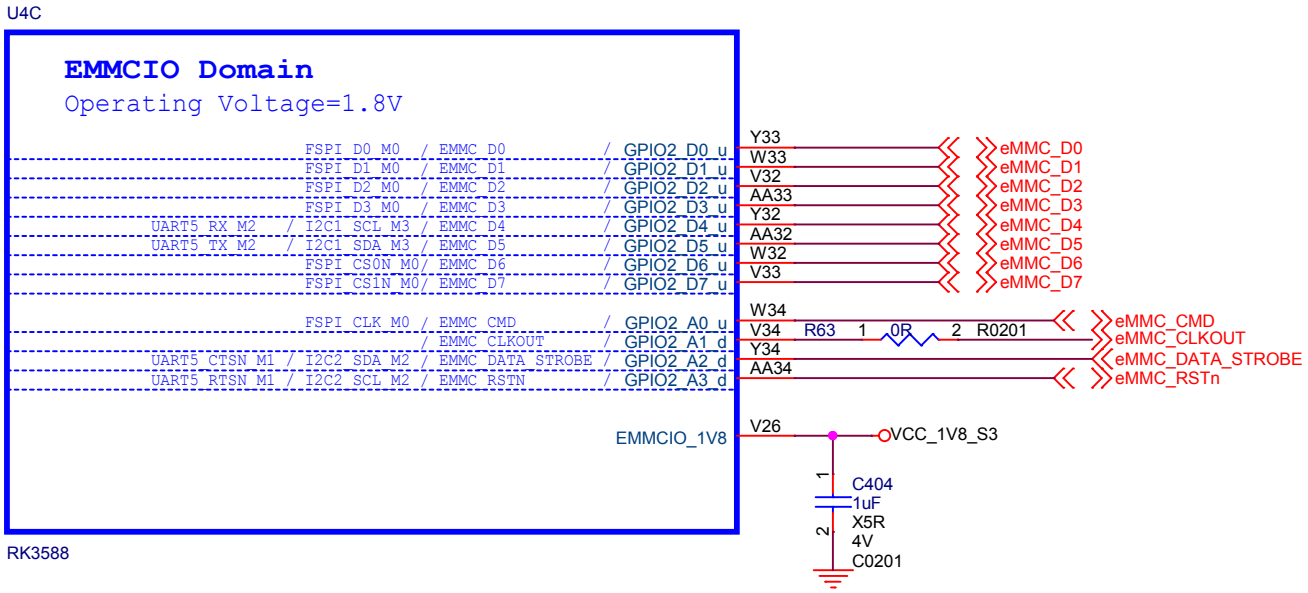
RK3588_F(PMUIO2)



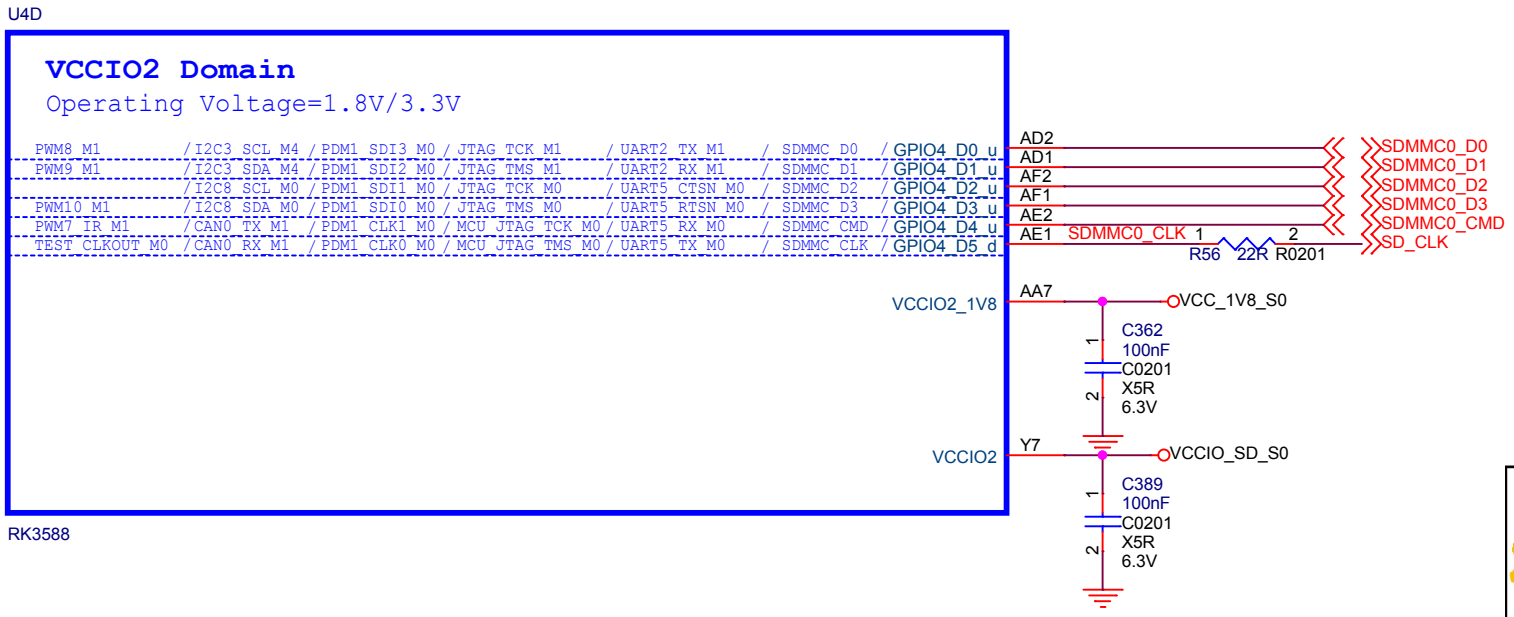
Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 MAX				
File:	02.RK3588_OSC/PLL/PMUIO				
Date:	Friday, July 25, 2025			Rev:	V1.1
Size:	A3	Sheet:	2	Of:	29


U4A
U4B
RK3588
DDR_PLL POWER
LP4/4X_CKE&LP5_CS & Reset Power
MEMORY INTERFACE POWER
CK Power
DIGITAL CORE POWER
DDR IO POWER
DDR FILTER
VDDQ_DDR_S0
VDDQ_DDR_CK_S0
VDDQ_DDR_CKE_S3
VDD_DDR_PLL_S0
VCCA_1V8_S0
Shenzhen Xunlong Software Co., Ltd
Project: OPi 5 MAX
File: 03.RK3588 DDR Controler
Date: Friday, July 25, 2025
Rev: V1.1
Size: A3
Sheet: 3
Of: 29

RK3588_C(EMMCIO Domain)



RK3588_D(VCCIO2 Domain)



Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 MAX				
File:	04.RK3588_Flash/SD Controller				
Date:	Friday, July 25, 2025			Rev:	V1.1
Size:	A3	Sheet:	4	Of:	29

RK3588_M(TYPEC/DP)

U4M

USB3.0 OTG/DP1.4 Alt of TYPEC0

USB:U3/Gen1----Controller0
DP:RBR/HBR/HBR2/HBR3

TYPEC0_SBU1/DP0_AUXP — AL15
TYPEC0_SBU2/DP0_AUXN — AM15
TYPEC0_SSRX1P/DP0_TX0P — AN13 — TYPEC0_SSRX1P
TYPEC0_SSRX1N/DP0_TX0N — AP13 — TYPEC0_SSRX1N
TYPEC0_SSTX1P/DP0_TX1P — AP14 — TYPEC0_SSTX1P
TYPEC0_SSTX1N/DP0_TX1N — AN14 — TYPEC0_SSTX1N
TYPEC0_SSRX2P/DP0_TX2P — AN15
TYPEC0_SSRX2N/DP0_TX2N — AP15
TYPEC0_SSTX2P/DP0_TX3P — AP16
TYPEC0_SSTX2N/DP0_TX3N — AN16
TYPEC0_DP0_REXT — AH16 — TYPEC0_REXT — R197 8.2K R0201 1%
TYPEC0_DP0_VDD_0V85 — AJ14
TYPEC0_DP0_VDDA_0V85 — AH14 — 340mA — VDDA_0V85_S0
C271 1uF X5R 6.3V C0201
C272 100nF X5R 6.3V C0201
TYPEC0_DP0_VDDH_1V8 — AG14 — 60mA — VCCA_1V8_S0
C291 1uF X5R 6.3V C0201
C289 100nF X5R 6.3V C0201

To USB3.0

USB3.0 OTG/DP1.4 Alt of TYPEC1

USB:U3/Gen1----Controller1
DP:RBR/HBR/HBR2/HBR3

TYPEC1_SBU1/DP1_AUXP — AL10
TYPEC1_SBU2/DP1_AUXN — AM10
TYPEC1_SSRX1P/DP1_TX0P — AN8 — TYPEC1_SSRX1P
TYPEC1_SSRX1N/DP1_TX0N — AP8 — TYPEC1_SSRX1N
TYPEC1_SSTX1P/DP1_TX1P — AP9 — TYPEC1_SSTX1P
TYPEC1_SSTX1N/DP1_TX1N — AN9 — TYPEC1_SSTX1N
TYPEC1_SSRX2P/DP1_TX2P — AN10
TYPEC1_SSRX2N/DP1_TX2N — AP10
TYPEC1_SSTX2P/DP1_TX3P — AP11
TYPEC1_SSTX2N/DP1_TX3N — AN11
TYPEC1_DP1_REXT — AG16 — TYPEC1_REXT — R200 8.2K R0201
TYPEC1_DP1_VDD_0V85 — AH13
TYPEC1_DP1_VDDA_0V85 — AJ13 — 340mA — VDDA_0V85_S0
C270 1uF X5R 6.3V C0201
C273 100nF X5R 6.3V C0201
TYPEC1_DP1_VDDH_1V8 — AG13 — 60mA — VCCA_1V8_S0
C292 1uF X5R 6.3V C0201
C293 100nF X5R 6.3V C0201

To USB3.0

Note:

RK3588

RK3588_L(USB2.0 HOST/OTG)

U4L

USB2.0 of TYPEC0 (OTG/HOST/DEVICE)

HS/FS/LS

Download Port
TYPEC0_USB20_OTG_DP — AL12 — TYPEC0_OTG_DP
TYPEC0_USB20_OTG_DM — AM12 — TYPEC0_OTG_DM
to USB3.0
TYPEC0_USB20_OTG_ID — AL14 — =1.8V
40K — TYPEC0_USB20_VBUSDET — AM14 — <=3.3V — TYPEC0_USB20_VBUSDET
TYPEC0_USB20_OTG0_REXT — AP12 — OTG0_REXT — R41 200R R0201

USB2.0 of TYPEC1 (OTG/HOST/DEVICE)

HS/FS/LS

TYPEC1_USB20_OTG_DP — AK9 — TYPEC1_OTG_DP
TYPEC1_USB20_OTG_DM — AL9 — TYPEC1_OTG_DM
to USB3.0
TYPEC1_USB20_OTG_ID — AK8
40K — TYPEC1_USB20_VBUSDET — AL8
TYPEC1_USB20_OTG1_REXT — AP7 — OTG1_REXT — R40 200R R0201

USB2.0 HOST0

HS/FS/LS

USB20_HOST0_DP — AK6 — USB20_HOST0_DP
USB20_HOST0_DM — AL6 — USB20_HOST0_DM
to USB2.0
USB20_HOST0_REXT — AG9 — HOST0_REXT — R201 200R R0201

USB2.0 HOST1

HS/FS/LS

USB20_HOST1_DP — AL7 — USB20_HOST1_DP
USB20_HOST1_DM — AM7 — USB20_HOST1_DM
to USB2.0
USB20_HOST1_REXT — AH9 — HOST1_REXT — R195 200R R0201

USB2.0 POWER

USB20_DVDD_0V75 — AH10 — 30mA — VDD_0V75_S0
C281 100nF X5R 6.3V C0201
USB20_AVDD_1V8 — AG11 — 65mA — VCCA_1V8_S0
C290 100nF X5R 6.3V C0201
USB20_AVDD_3V3 — AJ10 — 45mA — VCCA_3V3_S0
C267 100nF X5R 6.3V C0201

RK3588

Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 MAX				
File:	05.RK3588_USB30/USB20_Ctrl				
Date:	Friday, July 25, 2025			Rev:	V1.1
Size:	A3	Sheet:	5	Of:	29

RK3588_G(VCCIO1 Domain)

U4G

VCCIO1 Domain

Operating Voltage=1.8V

SPI4_MISO_M0	UART3_RX_M0		I2C3_SDA_M0			GPIO1_C0_z
SPI4_MOSI_M0	UART3_TX_M0		I2C3_SCL_M0			GPIO1_C1_z
SPI4_CLK_M0	UART3_RTSN	PWM3_IR_M2	I2C6_SDA_M1		I2S0_MCLK	GPIO1_C2_d
SPI4_CS0_M0	UART3_CTSN	PWM7_IR_M2	I2C6_SCL_M1		I2S0_SCLK	GPIO1_C3_d
SPI4_CS1_M0		PWM11_IR_M2	I2C2_SDA_M3	PDM0_CLK1_M0		GPIO1_C4_d
	UART4_RTSN		I2C2_SCL_M3		I2S0_LRCK	GPIO1_C5_d
		PWM15_IR_M2	I2C4_SDA_M4	PDM0_CLK0_M0		GPIO1_C6_d
	UART4_CTSN		I2C4_SCL_M4		I2S0_SDO0	GPIO1_C7_d
SPI1_MISO_M2	UART6_TX_M2		I2C7_SCL_M0		I2S0_SDO1	GPIO1_D0_d
SPI1_MOSI_M2	UART6_RX_M2		I2C7_SDA_M0	PDM0_SDI1_M0	I2S0_SDO2/I2S0_SDI3	GPIO1_D1_d
SPI1_CLK_M2	UART4_TX_M0	PWM0_M1	I2C1_SCL_M4	PDM0_SDI2_M0	I2S0_SDO3/I2S0_SDI2	GPIO1_D2_d
SPI1_CS0_M2	UART4_RX_M0	PWM1_M1	I2C1_SDA_M4	PDM0_SDI3_M0	I2S0_SDI1	GPIO1_D3_d
					I2S0_SDI0	GPIO1_D4_d
SPI1_CS1_M2				PDM0_SDI0_M0		GPIO1_D5_d

VCCIO1_1V8

G29 — I2C3_SDA_M0
G27 — I2C3_SCL_M0
F30 — I2S0_MCLK
E31 — I2S0_SCLK
E30 — CAM0_RST
D30 — I2S0_LRCK
D29 — LCD_BL_PWM1
E29 — I2S0_SDO0
F26 — I2C7_SCL_M0
F27 — I2C7_SDA_M0
F28 — LCD_PWREN_H
E28 — LCD_RESET
D28 — I2S0_SDI0
G26 — CAM0_PDN
G20 — VCC_1V8_S0

C545 100nF X5R 6.3V C0201

VCC_1V8_S0
R287 2.2K 5% R0201 — I2C3_SCL_M0
R286 2.2K 5% R0201 — I2C3_SDA_M0

VCC_1V8_S0
R285 2.2K 5% R0201 — I2C7_SCL_M0
R284 2.2K 5% R0201 — I2C7_SDA_M0

RK3588

RK3588_H(VCCIO3 Domain)

U4H

VCCIO3 Domain

Operating Voltage=1.8V

	UART6_RX_M0	FSPI_D0_M1	SDIO_D0_M0		GMAC0_RXD2	GPIO2_A6_u
	UART6_TX_M0	FSPI_D1_M1	SDIO_D1_M0		GMAC0_RXD3	GPIO2_A7_u
I2C8_SCL_M1	UART6_RTSN_M0	FSPI_D2_M1	SDIO_D2_M0		GMAC0_RXCLK	GPIO2_B0_u
I2C8_SDA_M1	UART6_CTSN_M0	FSPI_D3_M1	SDIO_D3_M0		GMAC0_RXD2	GPIO2_B1_u
I2C3_SCL_M3			SDIO_CMD_M0		GMAC0_RXD3	GPIO2_B2_u
I2C3_SDA_M3		FSPI_CLK_M1	SDIO_CLK_M0		GMAC0_TXCLK	GPIO2_B3_d
I2C4_SDA_M1	UART7_RX_M0	FSPI_CSON_M1	HDMI_TX1_SDA_M0		GMAC0_PTP_REFCLK	GPIO2_B4_u
I2C4_SCL_M1	UART7_TX_M0	FSPI_CSIN_M1	HDMI_TX1_SCL_M0		GMAC0_PPSTRIG	GPIO2_B5_u
I2C5_SCL_M4	UART1_RX_M0		I2S2_MCLK_M0		GMAC0_TXD0	GPIO2_B6_d
I2C5_SDA_M4	UART1_TX_M0		I2S2_SCLK_M0		GMAC0_TXD1	GPIO2_B7_d
I2C2_SDA_M1	UART1_RTSN_M0	SPI1_CLK_M0	I2S2_LRCK_M0		GMAC0_TXEN	GPIO2_C0_d
I2C2_SCL_M1	UART1_CTSN_M0	SPI1_MISO_M0			GMAC0_TXD0	GPIO2_C1_d
I2C6_SDA_M2	UART9_TX_M0	SPI1_MOSI_M0			GMAC0_RXD1	GPIO2_C2_d
I2C6_SCL_M2		SPI1_CS0_M0	I2S2_SDI_M0		ETH0_REFCLKO_25M	GPIO2_C3_d
TEST_CLKOUT_M1	UART9_RX_M0	SPI1_CS1_M0	HDMI_TX1_CEC_M0		GMAC0_PPSCLK	GPIO2_C4_d
					CLK32K_OUT1	GPIO2_C5_d
	UART7_RTSN_M0	SPI3_CS0_M0	PWM2_M2		GMAC0_RXDV_CRS	GPIO4_C2_d
I2C1_SCL_M1	PWM4_M1	SPI3_CS1_M0	I2S2_SDO_M0		GMAC0_MCLKINOUT	GPIO4_C3_d
I2C1_SDA_M1	UART9_RTSN_M0	SPI3_MISO_M0	PWM5_M2		GMAC0_MDC	GPIO4_C4_d
I2C0_SCL_M1	UART9_CTSN_M0	SPI3_MOSI_M0	PWM6_M2		GMAC0_MDIO	GPIO4_C5_d
I2C0_SDA_M1	UART7_CTSN_M0	SPI3_CLK_M0	PWM7_IR_M3		GMAC0_TXER	GPIO4_C6_d

VCCIO3_1V8

AC32 — SDIO_D0_M0_WIFI
AC31 — SDIO_D1_M0_WIFI
AE32 — SDIO_D2_M0_WIFI
AC33 — SDIO_D3_M0_WIFI
AC34 — SDIO_CMD_M0_WIFI
AE33 — SDIO_CLK_M0_WIFI
AB31 — UART7_RX_M0_BT
AB30 — UART7_TX_M0_BT
AD33 — PCIE_PWREN_H
AD34 — I2S2_SCLK_M0_BT
AE34 — I2S2_LRCK_M0_BT
AD32 — TP_INT_L
AD31 — TP_RST_L
AD30 — I2S2_SDI_M0_BT
AC30 — HDMITX1_CEC_M0
AE30 — WIFI_REG_ON
AE31 — UART7_RTSn_M0_BT
AF34 — I2S2_SDO_M0_BT
AB34 — BT_REG_ON
AB33 — HOST_WAKE_BT
AF33 — UART7_CTSn_M0_BT
Y26 — VCC_1V8_S3

C387 100nF X5R 6.3V C0201

RK3588

RK3588_U(SARADC/OTP)

U4U

SARADC

12-bit 1MS/s

SARADC_IN0_BOOT Recovery/
SARADC_IN1
SARADC_IN2
SARADC_IN3
SARADC_IN4
SARADC_IN5
SARADC_IN6
SARADC_IN7

AM16 BOOT_SARADC_IN0
AL16 SARADC_VIN1_KEY/RECOVERY
AK16 SARADC_VIN2_LCD_ID
AN17 SARADC_VIN3_HP_HOOK
AM17
AK15
AL17
AK17

R174 1 — 2 10K R0201 1% — VCCA_1V8_S0
C240 1 — 2 1nF C0201 X5R 50V
C93 1 — 2 1nF C0201 X5R 50V
C23 1 — 2 1nF C0201 X5R 50V

SARADC_AVDD_1V8
AH18 — 5mA — VCCA_1V8_S0
C287 100nF X5R 6.3V C0201

OTP

NC AD4 — Must floating

OTP_VDDOTP_0V75
AD3 — 1mA — VDD_0V75_S0
C320 100nF X5R 6.3V C0201

TSADC

TSADC_TEST_OUT_TS
AF18

RK3588

BOOT MODE CONFIG

VCCA_1V8_S0
R170 100K 1% R0201
BOOT_SARADC_IN0
R171 DNP R0201

TABLE 1

Item	Rup	Rdown	ADC	VOL	BOOT MODE
LEVEL1	DNP	100K	0	0V	USB (Maskrom mode)
LEVEL2	100K	20K	682	0.3V	SD Card-USB
LEVEL3	100K	51K	1365	0.6V	EMMC-USB
LEVEL4	100K	100K	2047	0.9V	FSPI M0-USB
LEVEL5	100K	200K	2730	1.2V	FSPI M1-USB
LEVEL6	100K	499K	3412	1.5V	FSPI M2-USB
LEVEL7	100K	DNP	4095	1.8V	FSPI_M2-FSPI_M1-FSPI_M0 -EMMC-SD Card-USB

Shenzhen Xunlong Software Co., Ltd

Project:	OPi 5 MAX				
File:	06.RK3588_SARADC/1.8V Only GPIO				
Date:	Friday, July 25, 2025			Rev:	V1.1
Size:	A3	Sheet:	6	Of:	29

RK3588_Q/R(MIPI_D/C_PHY0/1)

U4Q

MIPI D/C-PHY DSI_TX Port0

D-PHY:V2.0 4.5Gbps/Lane
C-PHY:V1.1 5.7Gbps/Trio

MIPI_DPHY0_TX_CLKP/MIPI_CPHY0_TX_TRIO1_C	AN26
MIPI_DPHY0_TX_CLKN/MIPI_CPHY0_TX_TRIO1_B	AP26
MIPI_DPHY0_TX_D0P/MIPI_CPHY0_TX_TRIO0_B	AN24
MIPI_DPHY0_TX_D0N/MIPI_CPHY0_TX_TRIO0_A	AP24
MIPI_DPHY0_TX_D1P/MIPI_CPHY0_TX_TRIO1_A	AN25
MIPI_DPHY0_TX_D1N/MIPI_CPHY0_TX_TRIO0_C	AP25
MIPI_DPHY0_TX_D2P/MIPI_CPHY0_TX_TRIO2_B	AN27
MIPI_DPHY0_TX_D2N/MIPI_CPHY0_TX_TRIO2_A	AP27
MIPI_DPHY0_TX_D3P/NO_USE_AN28	AN28
MIPI_DPHY0_TX_D3N/MIPI_CPHY0_TX_TRIO2_C	AP28

MIPI D/C-PHY CSI_RX Port0

D-PHY:V1.2 2.5Gbps/Lane
C-PHY:V1.1 5.7Gbps/Trio

Note:
此D-PHY使用受限!!!
详细情况请参阅硬件设计指南。
This D-PHY is limited!!!
Please refer to the hardware design guide for details.

MIPI_DPHY0_RX_CLKP/MIPI_CPHY0_RX_TRIO1_C	AN32
MIPI_DPHY0_RX_CLKN/MIPI_CPHY0_RX_TRIO1_B	AP31
MIPI_DPHY0_RX_D0P/MIPI_CPHY0_RX_TRIO0_B	AN29
MIPI_DPHY0_RX_D0N/MIPI_CPHY0_RX_TRIO0_A	AP29
MIPI_DPHY0_RX_D1P/MIPI_CPHY0_RX_TRIO1_A	AN30
MIPI_DPHY0_RX_D1N/MIPI_CPHY0_RX_TRIO0_C	AP30
MIPI_DPHY0_RX_D2P/MIPI_CPHY0_RX_TRIO2_B	AN33
MIPI_DPHY0_RX_D2N/MIPI_CPHY0_RX_TRIO2_A	AP32
MIPI_DPHY0_RX_D3P/NO_USE_AN34	AN34
MIPI_DPHY0_RX_D3N/MIPI_CPHY0_RX_TRIO2_C	AP33

MIPI_DPHY0_RX_CLKP/MIPI_CPHY0_RX_TRIO1_C	AN32	AP31	MIPI_DPHY0_RX_CLKN/MIPI_CPHY0_RX_TRIO1_B
MIPI_DPHY0_RX_D0P/MIPI_CPHY0_RX_TRIO0_B	AN29	AP29	MIPI_DPHY0_RX_D0N/MIPI_CPHY0_RX_TRIO0_A
MIPI_DPHY0_RX_D1P/MIPI_CPHY0_RX_TRIO1_A	AN30	AP30	MIPI_DPHY0_RX_D1N/MIPI_CPHY0_RX_TRIO0_C
MIPI_DPHY0_RX_D2P/MIPI_CPHY0_RX_TRIO2_B	AN33	AP32	MIPI_DPHY0_RX_D2N/MIPI_CPHY0_RX_TRIO2_A
MIPI_DPHY0_RX_D3P/NO_USE_AN34	AN34	AP33	MIPI_DPHY0_RX_D3N/MIPI_CPHY0_RX_TRIO2_C

MIPI D/C-PHY POWER

MIPI_D/C_PHY0_VREG	AF20	C304 1uF		C0201 X5R 4V	
MIPI_D/C_PHY0_VDD	AG20	C286 100nF X5R 6.3V C0201	C285 1uF X5R 6.3V C0201	130mA	VDD_0V75_S0
MIPI_D/C_PHY0_VDD_1V2	AH20	C278 1uF X5R 6.3V C0201		4mA	VDDA_1V2_S0
MIPI_D/C_PHY0_VDD_1V8	AJ20	C268 1uF X5R 6.3V C0201		35mA	VCC_1V8_S0

RK3588

U4R

MIPI D/C-PHY DSI_TX Port1

D-PHY:V2.0 4.5Gbps/Lane
C-PHY:V1.1 5.7Gbps/Trio

MIPI_DPHY1_TX_CLKP/MIPI_CPHY1_TX_TRIO1_C	AN20
MIPI_DPHY1_TX_CLKN/MIPI_CPHY1_TX_TRIO1_B	AP20
MIPI_DPHY1_TX_D0P/MIPI_CPHY1_TX_TRIO0_B	AN18
MIPI_DPHY1_TX_D0N/MIPI_CPHY1_TX_TRIO0_A	AP18
MIPI_DPHY1_TX_D1P/MIPI_CPHY1_TX_TRIO1_A	AN19
MIPI_DPHY1_TX_D1N/MIPI_CPHY1_TX_TRIO0_C	AP19
MIPI_DPHY1_TX_D2P/MIPI_CPHY1_TX_TRIO2_B	AN21
MIPI_DPHY1_TX_D2N/MIPI_CPHY1_TX_TRIO2_A	AP21
MIPI_DPHY1_TX_D3P /NO_USE_AN22	AN22
MIPI_DPHY1_TX_D3N/MIPI_CPHY1_TX_TRIO2_C	AP22

MIPI_DPHY1_TX_CLKP/MIPI_CPHY1_TX_TRIO1_C	AN20	AP20	MIPI_DPHY1_TX_CLKN/MIPI_CPHY1_TX_TRIO1_B
MIPI_DPHY1_TX_D0P/MIPI_CPHY1_TX_TRIO0_B	AN18	AP18	MIPI_DPHY1_TX_D0N/MIPI_CPHY1_TX_TRIO0_A
MIPI_DPHY1_TX_D1P/MIPI_CPHY1_TX_TRIO1_A	AN19	AP19	MIPI_DPHY1_TX_D1N/MIPI_CPHY1_TX_TRIO0_C
MIPI_DPHY1_TX_D2P/MIPI_CPHY1_TX_TRIO2_B	AN21	AP21	MIPI_DPHY1_TX_D2N/MIPI_CPHY1_TX_TRIO2_A
MIPI_DPHY1_TX_D3P /NO_USE_AN22	AN22	AP22	MIPI_DPHY1_TX_D3N/MIPI_CPHY1_TX_TRIO2_C

MIPI D/C-PHY CSI_RX Port1

D-PHY:V1.2 2.5Gbps/Lane
C-PHY:V1.1 5.7Gbps/Trio

Note:
此D-PHY使用受限!!!
详细情况请参阅硬件设计指南。
This D-PHY is limited!!!
Please refer to the hardware design guide for details.

MIPI_DPHY1_RX_CLKP/MIPI_CPHY1_RX_TRIO1_C	AK20
MIPI_DPHY1_RX_CLKN/MIPI_CPHY1_RX_TRIO1_B	AL20
MIPI_DPHY1_RX_D0P/MIPI_CPHY1_RX_TRIO0_B	AK18
MIPI_DPHY1_RX_D0N/MIPI_CPHY1_RX_TRIO0_A	AL18
MIPI_DPHY1_RX_D1P/MIPI_CPHY1_RX_TRIO1_A	AK19
MIPI_DPHY1_RX_D1N/MIPI_CPHY1_RX_TRIO0_C	AL19
MIPI_DPHY1_RX_D2P/MIPI_CPHY1_RX_TRIO2_B	AK21
MIPI_DPHY1_RX_D2N/MIPI_CPHY1_RX_TRIO2_A	AL21
MIPI_DPHY1_RX_D3P/NO_USE_AK22	AK22
MIPI_DPHY1_RX_D3N/MIPI_CPHY1_RX_TRIO2_C	AL22

MIPI_D/C_PHY1_VREG	AF19	C303 1uF		C0201 X5R 4V	
MIPI_D/C_PHY1_VDD	AG19	C284 1uF X5R 6.3V C0201		130mA	VDD_0V75_S0
MIPI_D/C_PHY1_VDD_1V2	AH19	C280 100nF X5R 6.3V C0201		4mA	VDDA_1V2_S0
MIPI_D/C_PHY1_VDD_1V8	AJ19	C269 1uF X5R 6.3V C0201		35mA	VCC_1V8_S0

RK3588

RK3588_P(MIPI_DPHY_CSI_RX_PHY)

U4P

MIPI DPHY CSI_RX Port0

MIPI V1.2/2.5Gbps

MIPI_CSI0_CLK0P	AJ33	AJ34	MIPI_CSI0_RX_CLK0P
MIPI_CSI0_CLK0N			MIPI_CSI0_RX_CLK0N
MIPI_CSI0_D0P	AG33	AG34	MIPI_CSI0_RX_D0P
MIPI_CSI0_D0N			MIPI_CSI0_RX_D0N
MIPI_CSI0_D1P	AH33	AH34	MIPI_CSI0_RX_D1P
MIPI_CSI0_D1N			MIPI_CSI0_RX_D1N
MIPI_CSI0_CLK1P	AM33	AM34	
MIPI_CSI0_CLK1N			
MIPI_CSI0_D2P	AK33	AK34	MIPI_CSI0_RX_D2P
MIPI_CSI0_D2N			MIPI_CSI0_RX_D2N
MIPI_CSI0_D3P	AL33	AL34	MIPI_CSI0_RX_D3P
MIPI_CSI0_D3N			MIPI_CSI0_RX_D3N

MIPI_CSI0_AVCC0V75	AB25	C343 100nF X5R 6.3V C0201	C363 1uF X5R 6.3V C0201	10mA	VDD_0V75_S0
MIPI_CSI0_AVCC1V8	AB26	C346 1uF X5R 6.3V C0201		3.3mA	VCC_1V8_S0

MIPI DPHY CSI_RX Port1

MIPI V1.2/2.5Gbps

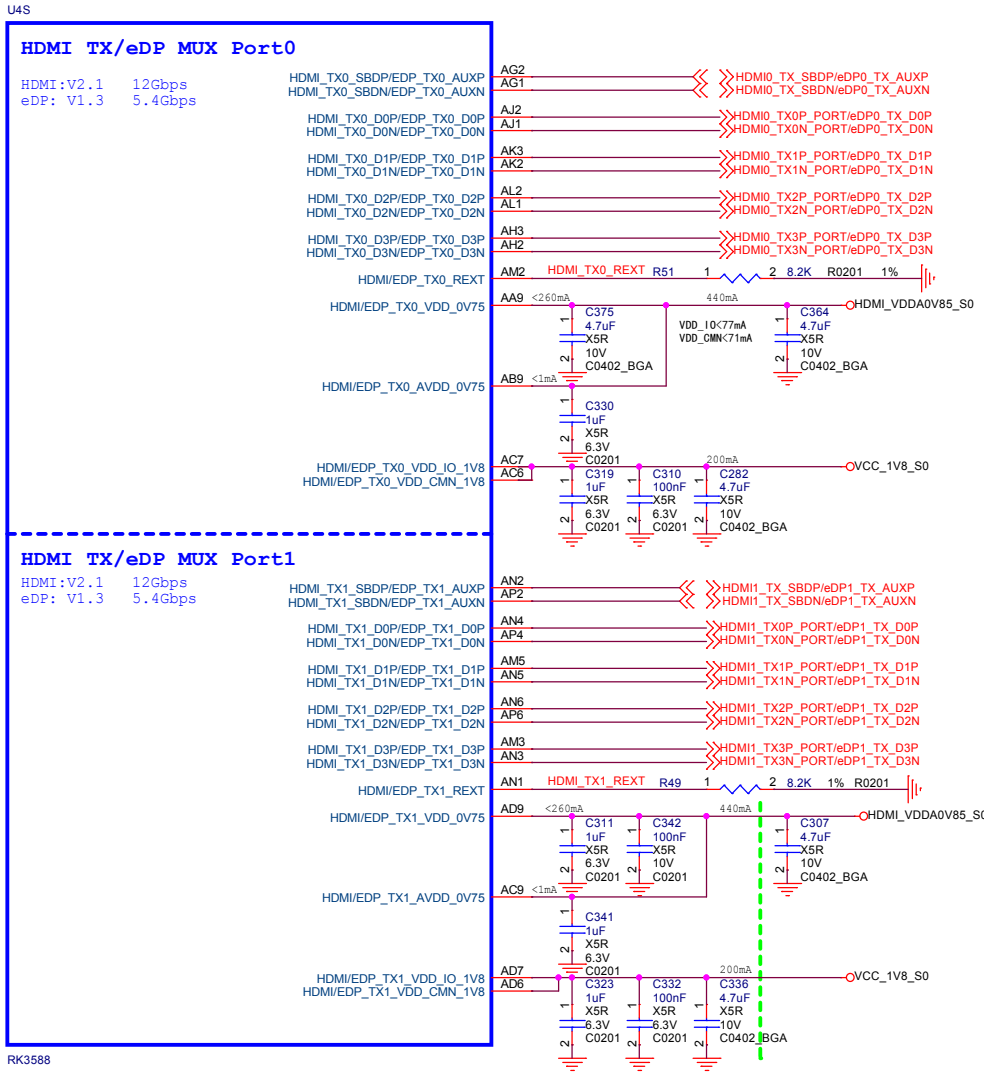
MIPI_CSI1_CLK0P	AJ31	AJ32	MIPI_CSI1_RX_CLK0P
MIPI_CSI1_CLK0N			MIPI_CSI1_RX_CLK0N
MIPI_CSI1_D0P	AG31	AG32	MIPI_CSI1_RX_D0P
MIPI_CSI1_D0N			MIPI_CSI1_RX_D0N
MIPI_CSI1_D1P	AH31	AH32	MIPI_CSI1_RX_D1P
MIPI_CSI1_D1N			MIPI_CSI1_RX_D1N
MIPI_CSI1_CLK1P	AM31	AM32	
MIPI_CSI1_CLK1N			
MIPI_CSI1_D2P	AK31	AK32	MIPI_CSI1_RX_D2P
MIPI_CSI1_D2N			MIPI_CSI1_RX_D2N
MIPI_CSI1_D3P	AL31	AL32	MIPI_CSI1_RX_D3P
MIPI_CSI1_D3N			MIPI_CSI1_RX_D3N

MIPI_CSI1_AVCC0V75	AA25	C361 1uF X5R 6.3V C0201		10mA	VDD_0V75_S0
MIPI_CSI1_AVCC1V8	AA26	C385 100nF X5R 6.3V C0201		3.3mA	VCC_1V8_S0

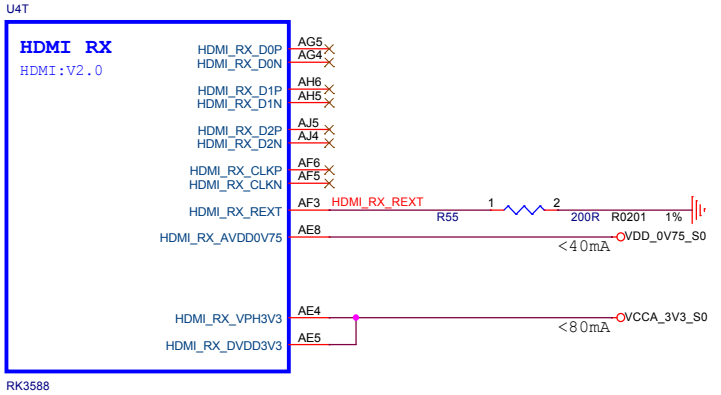
RK3588

Shenzhen Xunlong Software Co.,Ltd			
Project:	OPi 5 MAX		
File:	07.RK3588_MIPI Interface		
Date:	Friday, July 25, 2025	Rev:	V1.1
Size:	A3 Sheet: 7	Of:	29

RK3588_S(HDMI2.1 TX)



RK3588_T(HDMI20 RX)



Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 MAX				
File:	08.RK3588_HDMI/eDP Interface				
Date:	Friday, July 25, 2025			Rev:	V1.1
Size:	A3	Sheet:	8	Of:	29

RK3588_N(PCIE20)

U4N

PCIE20/SATA30 Mux0

PCIe20x1_2
1Lane(RC)--4(1L2)

SATA30 HOST
Controller0

MUX

input or output

PCIE20_0_REFCLKP
PCIE20_0_REFCLKN
L32
L33
PCIE20_0_REFCLKp
PCIE20_0_REFCLKN

PCIE20_0_TXP/SATA30_0_TXP
PCIE20_0_TXN/SATA30_0_TXN
M34
M33
PCIE20_0_TXP
PCIE20_0_TXN

PCIE20_0_RXP/SATA30_0_RXP
PCIE20_0_RXN/SATA30_0_RXN
N33
N34
PCIE20_0_RXP
PCIE20_0_RXN

PCIE20_SATA30_0_AVDD_0V85
M28
140mA
VDDA_0V85_S0
C523
10uF
X5R
4V
C0402_BGA

PCIE20_SATA30_0_AVDD_1V8
M27
270mA
VCCA_1V8_S0
C490
1uF
X5R
4V
C0201

PCIE20:Gen1/Gen2
SATA30:Gen1/Gen2/Gen3

PCIE20/SATA30 Mux1

PCIe30x1_0
1Lane(RC)--2(1L0)

SATA30 HOST
Controller1

MUX

input or output

PCIE20_1_REFCLKP
PCIE20_1_REFCLKN
H32
H33
PCIE20_1_REFCLKP
PCIE20_1_REFCLKN

PCIE20_1_TXP/SATA30_1_TXP
PCIE20_1_TXN/SATA30_1_TXN
K33
K34
PCIE20_1_TXP
PCIE20_1_TXN

PCIE20_1_RXP/SATA30_1_RXP
PCIE20_1_RXN/SATA30_1_RXN
J33
J34
PCIE20_1_RXP
PCIE20_1_RXN

PCIE20_SATA30_1_AVDD_0V85
L28
140mA
VDDA_0V85_S0
C488
100nF
X5R
6.3V
C0201

PCIE20_SATA30_1_AVDD_1V8
L27
270mA
VCCA_1V8_S0
C502
100nF
X5R
6.3V
C0201

PCIE20:Gen1/Gen2
SATA30:Gen1/Gen2/Gen3

PCIE20/SATA30/USB30 HOST Mux2

PCIe30x1_1
1Lane(RC)--3(1L1)

SATA30 HOST
Controller2

USB30 HOST
Controller2

MUX

input or output

PCIE20_2_REFCLKP
PCIE20_2_REFCLKN
G31
G30
PCIE20_2_REFCLKP
PCIE20_2_REFCLKN

PCIE20_2_TXP/SATA30_2_TXP/USB30_2_SSTXP
PCIE20_2_TXN/SATA30_2_TXN/USB30_2_SSTXN
H30
H29
PCIE20_2_TXP
PCIE20_2_TXN

PCIE20_2_RXP/SATA30_2_RXP/USB30_2_SSRXP
PCIE20_2_RXN/SATA30_2_RXN/USB30_2_SSRXN
J31
J30
PCIE20_2_RXP
PCIE20_2_RXN

PCIE20_SATA30_USB30_2_AVDD_0V85
K28
140mA
VDDA_0V85_S0
C498
1uF
X5R
4V
C0201

PCIE20_SATA30_USB30_2_AVDD_1V8
K27
270mA
VCCA_1V8_S0
C512
1uF
X5R
4V
C0201

PCIE20:Gen1/Gen2
USB 30:Gen1
SATA30:Gen1/Gen2/Gen3

RK3588

RK3588_O(PCIE30)

U4O

PCIE30X4

PCIe30X4
4Lane(DM)--0(4L)

PCIe30x1_0
1Lane(RC)--2(1L0)

PCIE20/SATA30
MUX1

PCIe30x2
2Lane(RC)--1(2L)

PCIe30x1_1
1Lane(RC)--3(1L1)

PCIE20/SATA30/USB30
MUX2

PCIE30:
PCIe1.1(2.5Gbps)
PCIe2.1(5Gbps)
PCIe3.0(8Gbps)

PCIE30 PORT0

input
PCIE30_PORT0_REF_CLKP
PCIE30_PORT0_REF_CLKN
E33
E34
PCIE30_PORT0_REFCLKP_IN
PCIE30_PORT0_REFCLKN_IN

Lane0
PCIE30_PORT0_TX0P
PCIE30_PORT0_TX0N
D32
D33
PCIE30_PORT0_TX0P
PCIE30_PORT0_TX0N

PCIE30_PORT0_RX0P
PCIE30_PORT0_RX0N
G33
G34
PCIE30_PORT0_RX0P
PCIE30_PORT0_RX0N

Lane1
PCIE30_PORT0_TX1P
PCIE30_PORT0_TX1N
C33
C34
PCIE30_PORT0_TX1P
PCIE30_PORT0_TX1N

PCIE30_PORT0_RX1P
PCIE30_PORT0_RX1N
F32
F33
PCIE30_PORT0_RX1P
PCIE30_PORT0_RX1N

PCIE30_PORT0_RESREF
B34
PCIE30_0_REF
R256
R0201
1
2
200R
1%

PCIE30_PORT0_AVDD0V75
G24
230mA
VDD_0V75_S0
C530
1uF
X5R
6.3V
C0201
C416
4.7uF
X5R
10V
C0402_BGA

PCIE30_PORT0_AVDD1V8
G23
210mA
VCCA_1V8_S0
C541
1uF
X5R
6.3V
C0201
C594
4.7uF
X5R
10V
C0402_BGA

PCIE30 PORT1

input
PCIE30_PORT1_REF_CLKP
PCIE30_PORT1_REF_CLKN
A28
B28
PCIE30_PORT1_REFCLKP_IN
PCIE30_PORT1_REFCLKN_IN

Lane0
PCIE30_PORT1_TX0P
PCIE30_PORT1_TX0N
B30
A30
PCIE30_PORT1_TX2P
PCIE30_PORT1_TX2N

PCIE30_PORT1_RX0P
PCIE30_PORT1_RX0N
B32
A32
PCIE30_PORT1_RX2P
PCIE30_PORT1_RX2N

Lane1
PCIE30_PORT1_TX1P
PCIE30_PORT1_TX1N
C29
B29
PCIE30_PORT1_TX3P
PCIE30_PORT1_TX3N

PCIE30_PORT1_RX1P
PCIE30_PORT1_RX1N
C31
B31
PCIE30_PORT1_RX3P
PCIE30_PORT1_RX3N

PCIE30_PORT1_RESREF
A33
PCIE30_1_REF
R258
R0201
1
2
200R
1%

PCIE30_PORT1_AVDD0V75
H24
230mA
VDD_0V75_S0
C540
1uF
X5R
6.3V
C0201
C417
4.7uF
X5R C0402_BGA
10V

PCIE30_PORT1_AVDD1V8
H23
210mA
VCCA_1V8_S0
C531
100nF
X5R
6.3V
C0201
C587
4.7uF
X5R
10V
C0402_BGA

RK3588

Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 MAX				
File:	09.RK3588_PCIE30/PCIE20/SATA30				
Date:	Friday, July 25, 2025			Rev:	V1.1
Size:	A3	Sheet:	9	Of:	29

RK3588_J(VCCIO5 Domain)

U4J

VCCIO5 Domain

Operating Voltage=1.8V/3.3V

PWM10_M0	SPI4_MISO_M1		I2C6_SDA_M4	FSPI_D0_M2	I2S3_MCLK	SDIO_D0_M1	GMAC1_TXD2	GPIO3_A0_u
AUDDSM_LN	SPI4_MOSI_M1	PWM11_IR_M0	I2C6_SCL_M4	FSPI_D1_M2	I2S3_SCLK	SDIO_D1_M1	GMAC1_TXD3	GPIO3_A1_u
AUDDSM_LP	SPI4_CLK_M1	UART8_TX_M1		FSPI_D2_M2	I2S3_LRCK	SDIO_D2_M1	GMAC1_RXD2	GPIO3_A2_u
AUDDSM_RN	SPI4_CS0_M1	UART8_RX_M1		FSPI_D3_M2	I2S3_SDO	SDIO_D3_M1	GMAC1_RXD3	GPIO3_A3_u
AUDDSM_RP	SPI4_CS1_M1	UART8_RTSN_M1			I2S3_SDI	SDIO_CMD_M1	GMAC1_TXCLK	GPIO3_A4_d
	MIPI_CAMERA0_CLK_M1	UART8_CTSN_M1	I2C4_SDA_M0	FSPI_CLK_M2		SDIO_CLK_M1	GMAC1_RXCLK	GPIO3_A5_d
	MIPI_CAMERA1_CLK_M1		I2C4_SCL_M0				ETH1_REFCLKO_25M	GPIO3_A6_d
PWM8_M0	MIPI_CAMERA2_CLK_M1						GMAC1_RXD0	GPIO3_A7_u
PWM9_M0	MIPI_CAMERA3_CLK_M1						GMAC1_RXD1	GPIO3_B0_u
PWM2_M1	MIPI_CAMERA4_CLK_M1	UART2_TX_M2					GMAC1_RXDV_CRS	GPIO3_B1_d
PWM3_IR_M1		UART2_RX_M2			I2S2_SDI_M1		GMAC1_TXER	GPIO3_B2_d
		UART2_RTSN			I2S2_SDO_M1		GMAC1_TXD0	GPIO3_B3_u
		UART2_CTSN			I2S2_MCLK_M1		GMAC1_TXD1	GPIO3_B4_u
PWM12_M0	CAN1_RX_M0	UART3_TX_M1			I2S2_SCLK_M1		GMAC1_TXEN	GPIO3_B5_u
PWM13_M0	CAN1_TX_M0	UART3_RX_M1			I2S2_LRCK_M1		GMAC1_MCLKINOUT	GPIO3_B6_d
			I2C3_SCL_M1	SPI1_MOSI_M1	HDMI_TX1_HPD_M1		GMAC1_PTP_REF_CLK	GPIO3_B7_d
		UART7_TX_M1	I2C3_SDA_M1	SPI1_MISO_M1			GMAC1_PPSTRIG	GPIO3_C0_d
	PCIE30X2_BUTTON_RSTN	UART7_RX_M1		SPI1_CLK_M1			GMAC1_PPSCLK	GPIO3_C1_d
PWM14_M0		UART7_RTSN_M1	I2C8_SCL_M4	SPI1_CS0_M1		MIPI_TE0	GMAC1_MDC	GPIO3_C2_d
PWM15_IR_M0		UART7_CTSN_M1	I2C8_SDA_M4	SPI1_CS1_M1		MIPI_TE1	GMAC1_MDIO	GPIO3_C3_d
CAN2_RX_M0	PCIE30X4_CLKREQN_M2	UART5_TX_M1	FSPI_CS0N_M2	SPI3_CS0_M3	HDMI_TX1_CEC_M2		CIF_D8	GPIO3_C4_u
CAN2_TX_M0	PCIE30X4_WAKEN_M2	UART5_RX_M1	FSPI_CS1N_M2	SPI3_CS1_M3	HDMI_TX1_SDA_M1		CIF_D9	GPIO3_C5_u
	PCIE30X4_PERSTN_M2			SPI3_MISO_M3	HDMI_TX1_SCL_M1		CIF_D10	GPIO3_C6_u
	PCIE20X1_2_CLKREQN_M0		I2C5_SCL_M0	SPI3_MOSI_M3	HDMI_TX0_SCL_M2		CIF_D11	GPIO3_C7_u
PWM8_M2	PCIE20X1_2_WAKEN_M0	UART4_RX_M1	I2C5_SDA_M0	SPI3_CLK_M3	HDMI_TX0_SDA_M2		CIF_D12	GPIO3_D0_u
PWM9_M2	PCIE20X1_2_PERSTN_M0	UART4_TX_M1		SPI0_MISO_M3	HDMI_RX_CEC_M1		CIF_D13	GPIO3_D1_d
	PCIE30X2_CLKREQN_M2	UART9_RTSN_M2	I2C7_SCL_M2	SPI0_MOSI_M3	HDMI_RX_SCL_M1		CIF_D14	GPIO3_D2_d
PWM10_M2	PCIE30X2_WAKEN_M2	UART9_CTSN_M2	I2C7_SDA_M2	SPI0_CLK_M3	HDMI_RX_SDA_M1		CIF_D15	GPIO3_D3_d
	PCIE30X2_PERSTN_M2	UART9_RX_M2		SPI0_CS0_M3	HDMI_RX_HPDOUT_M1	HDMI_TX0_HPD_M1	MCU_JTAG_TCK_M1	GPIO3_D4_d
PWM11_IR_M3	PCIE30X4_BUTTON_RSTN	UART9_TX_M2		SPI0_CS1_M3	DP1_HPDIN_M0		MCU_JTAG_TMS_M1	GPIO3_D5_d

AA29	FSPI_D0	GPIO3_A0_u
AA30	FSPI_D1	GPIO3_A1_u
AD27	FSPI_D2	GPIO3_A2_u
AE27	FSPI_D3	GPIO3_A3_u
AD28		GPIO3_A4_d
AH30	FSPI_CLK	GPIO3_A5_d
AH27	CAM0_CLK	GPIO3_A6_d
AG29	CAM1_CLK	GPIO3_A7_u
AG28	CAM2_CLK	GPIO3_B0_u
AH29	CAM1_RST	GPIO3_B1_d
AE28	CAM2_RST	GPIO3_B2_d
AC28	CAM1_PDN	GPIO3_B3_u
AC29	CAM2_PDN	GPIO3_B4_u
AD29	GPIO3_B5	GPIO3_B5_u
AE29	GPIO3_B6	GPIO3_B6_d
AA28	GPIO3_B7	GPIO3_B7_d
Y29	GPIO3_C0	GPIO3_C0_d
Y27	GPIO3_C1	GPIO3_C1_d
Y31	GPIO3_C2	GPIO3_C2_d
Y30	MIPI_TE1	GPIO3_C3_d
AH26	FSPI_CS0	GPIO3_C4_u
AH25	HDMITX1_SDA_M1	GPIO3_C5_u
AG26	HDMITX1_SCL_M1	GPIO3_C6_u
AJ24	PCIE20_1_2_CLKREQn_M0_L	GPIO3_C7_u
AH24	PCIE20_1_2_WAKEn_M0_L	GPIO3_D0_u
AG23	FAN_CTL_H	GPIO3_D1_d
AG25	HP_DET_L	GPIO3_D2_d
AG24		GPIO3_D3_d
AA27	PCIE20_1_2_PERSTN_L	GPIO3_D4_d
AB28	USB_HOST_PWREN	GPIO3_D5_d

VCCIO5_1V8 — W25 — C403 100nF X5R 6.3V C0201 — VCC_1V8_S0

VCCIO5 — W26 — C405 100nF X5R 6.3V C0201 — VCC_1V8_S0

RK3588

RK3588_K(VCCIO6 Domain)

U4K

VCCIO6 Domain

Operating Voltage=1.8V/3.3V

	SPI0_MISO_M1	UART9_RTSN_M1			I2S1_MCLK_M0	PCIE30X1_1_CLKREQN_M1	BT1120_D0	CIF_D0	GPIO4_A0_d
	SPI0_MOSI_M1	UART9_CTSN_M1			I2S1_SCLK_M0	PCIE30X1_1_WAKEN_M1	BT1120_D1	CIF_D1	GPIO4_A1_d
	SPI0_CLK_M1				I2S1_LRCK_M0	PCIE30X1_1_PERSTN_M1	BT1120_D2	CIF_D2	GPIO4_A2_d
		UART0_TX_M2				PCIE30X1_0_CLKREQN_M1	BT1120_D3	CIF_D3	GPIO4_A3_d
	SPI2_MISO_M1	UART0_RX_M2	I2C3_SCL_M2			PCIE30X1_0_WAKEN_M1	BT1120_D4	CIF_D4	GPIO4_A4_d
	SPI2_MOSI_M1	UART3_TX_M2	I2C3_SDA_M2		I2S1_SDI0_M0	PCIE30X1_0_PERSTN_M1	BT1120_D5	CIF_D5	GPIO4_A5_d
	SPI2_CLK_M1	UART3_RX_M2	I2C5_SCL_M2		I2S1_SDI1_M0	PCIE30X2_CLKREQN_M1	BT1120_D6	CIF_D6	GPIO4_A6_d
	SPI2_CS0_M1		I2C5_SDA_M2		I2S1_SDI2_M0	PCIE30X2_WAKEN_M1	BT1120_D7	CIF_D7	GPIO4_A7_d
	SPI2_CS1_M1	UART8_TX_M0	I2C6_SDA_M3		I2S1_SDI3_M0	PCIE30X2_PERSTN_M1	BT1120_CLKOUT	CIF_CLKIN	GPIO4_B0_d
SATA2_ACT_LED_M0	SPDIF1_TX_M1	SPI0_CS1_M1	UART8_RX_M0	I2C6_SCL_M3		I2S1_SDO0_M0	PCIE30X1_0_BUTTON_RSTN		MIPI_CAMERA5_CLK_M0	GPIO4_B1_u
CAN1_RX_M1	PWM14_M1	SPI0_CS0_M1	UART8_RTSN_M0	I2C7_SCL_M3		I2S1_SDO1_M0	PCIE30X1_1_BUTTON_RSTN	BT1120_D8	CIF_HREF	GPIO4_B2_u
CAN1_TX_M1	PWM15_IR_M1		UART8_CTSN_M0	I2C7_SDA_M3		I2S1_SDO2_M0	PCIE20X1_2_BUTTON_RSTN	BT1120_D9	CIF_VSYNC	GPIO4_B3_u
SPDIF0_TX_M1	PWM11_IR_M1	DP0_HPDIN_M0	UART9_TX_M1			I2S1_SDO3_M0	PCIE30X4_CLKREQN_M1	BT1120_D10	CIF_CLKOUT	GPIO4_B4_u
SATA1_ACT_LED_M0	PWM12_M1	SPI3_MISO_M1	UART9_RX_M1			HDMI_RX_CEC_M0	PCIE30X4_WAKEN_M1	BT1120_D11		GPIO4_B5_d
SATA0_ACT_LED_M0	PWM13_M1	SPI3_MOSI_M1		I2C5_SCL_M1		HDMI_RX_HPDOUT_M0	PCIE30X4_PERSTN_M1	BT1120_D12		GPIO4_B6_d
		SPI3_CLK_M1		I2C5_SDA_M1		HDMI_TX0_SCL_M0	PCIE20X1_2_CLKREQN_M1	BT1120_D13		GPIO4_B7_u
		SPI3_CS0_M1		I2C8_SCL_M3		HDMI_TX0_SDA_M0	PCIE20X1_2_WAKEN_M1	BT1120_D14		GPIO4_C0_u
SPDIF1_TX_M2	PWM6_M1	SPI3_CS1_M1		I2C8_SDA_M3		HDMI_TX0_CEC_M0	PCIE20X1_2_PERSTN_M1	BT1120_D15		GPIO4_C1_d

AK30	PCIE30x1_1_CLKREQn_M1_L	GPIO4_A0_d
AL30	PCIE30x1_1_WAKEn_M1_L	GPIO4_A1_d
AM29	PCIE30x1_1_PERSTn_M1_L	GPIO4_A2_d
AL29	PCIE30x1_0_CLKREQn_M1_L	GPIO4_A3_d
AL28	PCIE30x1_0_WAKEn_M1_L	GPIO4_A4_d
AK27	PCIE30x1_0_PERSTn_M1_L	GPIO4_A5_d
AL27	GPIO4_A6	GPIO4_A6_d
AM27	GPIO4_A7	GPIO4_A7_d
AK26	PHONE_CTL	GPIO4_B0_d
AL24	HDMI0_TX_ON_H	GPIO4_B1_u
AK25	HDMI1_TX_ON_H	GPIO4_B2_u
AM25	USB_OTG_PWREN	GPIO4_B3_u
AL26	PCIE30X4_CLKREQn_M1_L	GPIO4_B4_u
AJ26	PCIE30X4_WAKEn_M1_L	GPIO4_B5_d
AJ27	PCIE30X4_PERSTn_M1_L	GPIO4_B6_d
AJ28	HDMITX0_SCL_M0	GPIO4_B7_u
AJ25	HDMITX0_SDA_M0	GPIO4_C0_u
AK24	HDMITX0_CEC_M0	GPIO4_C1_d

VCCIO6_1V8 — AC25 — C327 100nF X5R 6.3V C0201 — VCC_1V8_S0

VCCIO6 — AC26 — C326 100nF X5R 6.3V C0201 — VCC_3V3_S0

RK3588

RK3588_I(VCCIO4 Domain)

U4I

VCCIO4 Domain

Operating Voltage=1.8V/3.3V

		SPI4_MISO_M2	PCIE30X1_1_CLKREQN_M2	DP0_HPDIN_M2		I2C2_SDA_M4	UART6_RX_M1	GPIO1_A0_d
SATA1_ACT_LED_M1		SPI4_MOSI_M2	PCIE30X1_1_WAKEN_M2	DP1_HPDIN_M2		I2C2_SCL_M4	UART6_TX_M1	GPIO1_A1_d
	PWM0_M2	SPI4_CLK_M2			VOP_POST_EMPTY	I2C4_SDA_M3	UART6_RTSN_M1	GPIO1_A2_d
	PWM1_M2	SPI4_CS0_M2			HDMI_TX1_SDA_M2	I2C4_SCL_M3	UART6_CTSN_M1	GPIO1_A3_d
		SPI2_MISO_M0			HDMI_TX1_SCL_M2			GPIO1_A4_d
		SPI2_MOSI_M0			HDMI_TX0_HPD_M0			GPIO1_A5_d
		SPI2_CLK_M0			HDMI_TX1_HPD_M0			GPIO1_A6_d
	PWM3_IR_M3	SPI2_CS0_M0	PCIE30X1_1_PERSTN_M2			PDM1_SDI0_M1		GPIO1_A7_u
		SPI2_CS1_M0	PCIE30X4_CLKREQN_M3			PDM1_SDI1_M1		GPIO1_B0_u
		SPI0_MISO_M2	PCIE30X4_WAKEN_M3			PDM1_SDI2_M1		GPIO1_B1_d
		SPI0_MOSI_M2	PCIE30X4_PERSTN_M3			PDM1_SDI3_M1	UART4_RX_M2	GPIO1_B2_d
SATA0_ACT_LED_M1		SPI0_CLK_M2	PCIE30X1_0_WAKEN_M2			PDM1_CLK1_M1	UART4_TX_M2	GPIO1_B3_d
		SPI0_CS0_M2	PCIE30X1_0_PERSTN_M2			PDM1_CLK0_M1	UART7_RX_M2	GPIO1_B4_u
		SPI0_CS1_M2	PCIE30X1_0_CLKREQN_M2				UART7_TX_M2	GPIO1_B5_u
	HDMI_RX_HPDOUT_M2	SPDIF0_TX_M0	PCIE30X2_WAKEN_M3		MIPI_CAMERA1_CLK_M0	I2C5_SCL_M3	UART1_TX_M1	GPIO1_B6_u
SATA2_ACT_LED_M1	HDMI_RX_CEC_M2	PWM13_M2	SPDIF1_TX_M0	PCIE30X2_PERSTN_M3	MIPI_CAMERA2_CLK_M0	I2C5_SDA_M3	UART1_RX_M1	GPIO1_B7_u
	HDMI_RX_SCL_M2	PWM14_M2			MIPI_CAMERA3_CLK_M0	I2C8_SCL_M2	UART1_RTSN_M1	GPIO1_D6_u
	HDMI_RX_SDA_M2	PWM15_IR_M3		PCIE30X2_CLKREQN_M3	MIPI_CAMERA4_CLK_M0	I2C8_SDA_M2	UART1_CTSN_M1	GPIO1_D7_u

A24	GPIO1_A0	GPIO1_A0_d
A25	GPIO1_A1	GPIO1_A1_d
A26	GPIO1_A2	GPIO1_A2_d
A27	GPIO1_A3	GPIO1_A3_d
B25	GPIO1_A4	GPIO1_A4_d
B26	HDMITX0_HPDIN_M0	GPIO1_A5_d
C24	HDMITX1_HPDIN_M0	GPIO1_A6_d
C25	GPIO1_A7/PWM3_IR_M3	GPIO1_A7_u
C27	GPIO1_B0	GPIO1_B0_u
D25	GPIO1_B1	GPIO1_B1_d
D26	GPIO1_B2	GPIO1_B2_d
D27	GPIO1_B3	GPIO1_B3_d
E24	GPIO1_B4	GPIO1_B4_u
E25	GPIO1_B5	GPIO1_B5_u
E26	GPIO1_B6	GPIO1_B6_u
E27	GPIO1_B7	GPIO1_B7_u
F24	GPIO1_D6	GPIO1_D6_u
F25	GPIO1_D7	GPIO1_D7_u

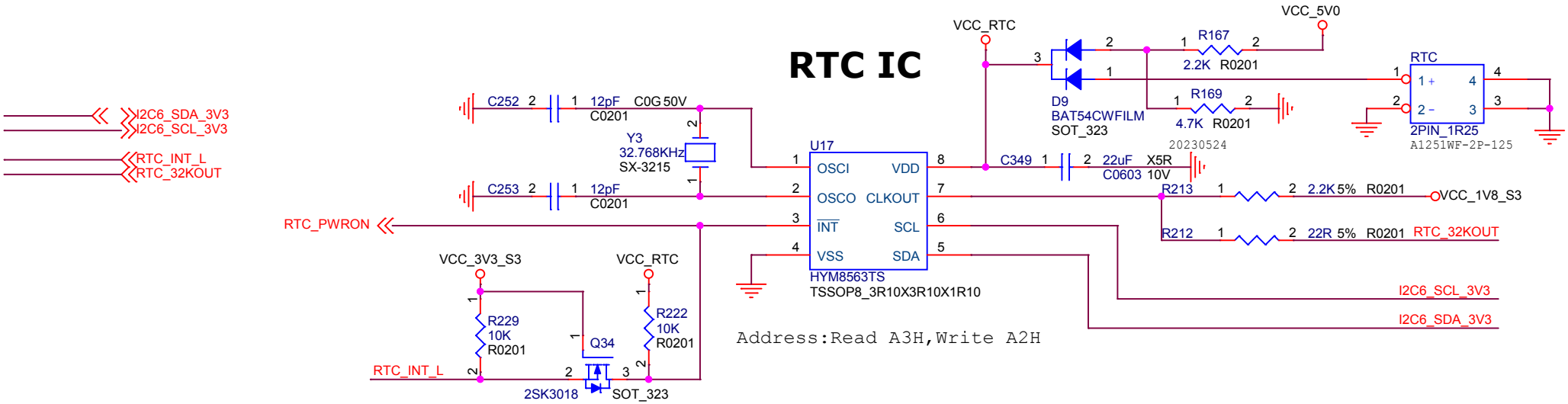
VCCIO4_1V8 — H20 — C543 100nF X5R 6.3V C0201 — VCC_1V8_S0

VCCIO4 — H21 — C534 100nF X5R 6.3V C0201 — VCC_3V3_S0

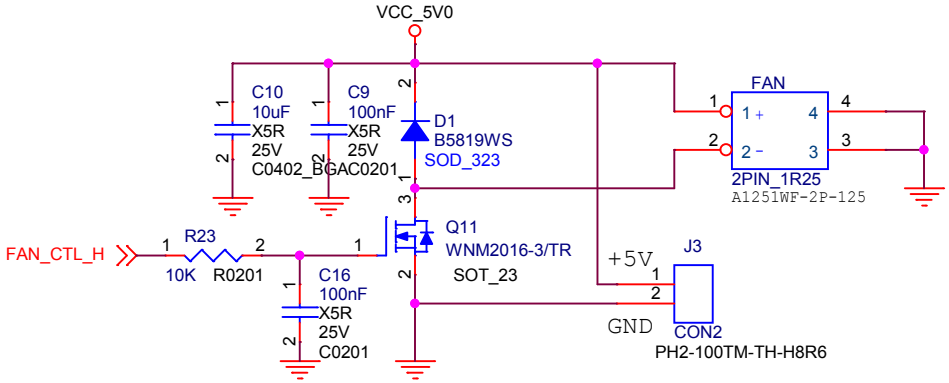
RK3588

Shenzhen Xunlong Software Co., Ltd		
Project:	OPi 5 MAX	
File:	10.RK3588_1.8V/ 3.3V GPIO	
Date:	Friday, July 25, 2025	Rev: V1.1
Size:	A3 Sheet: 10	Of: 29

RTC IC

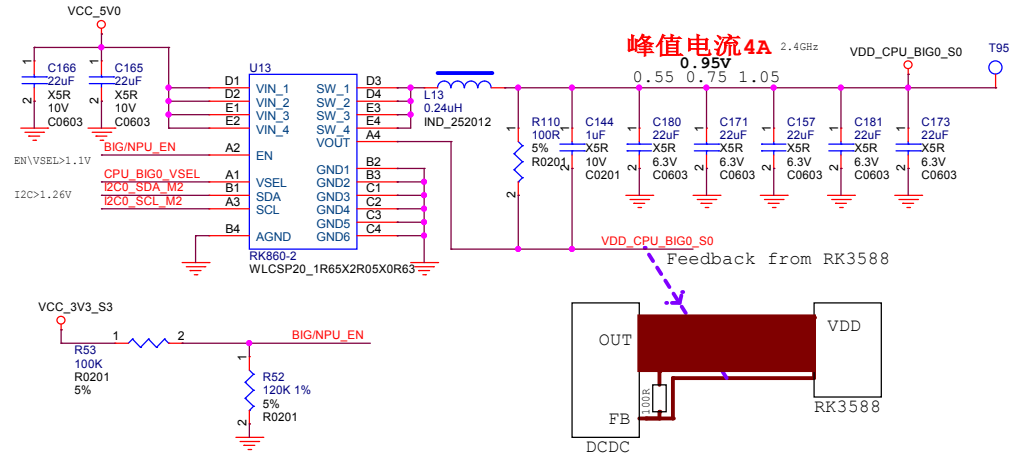


FAN_POWER

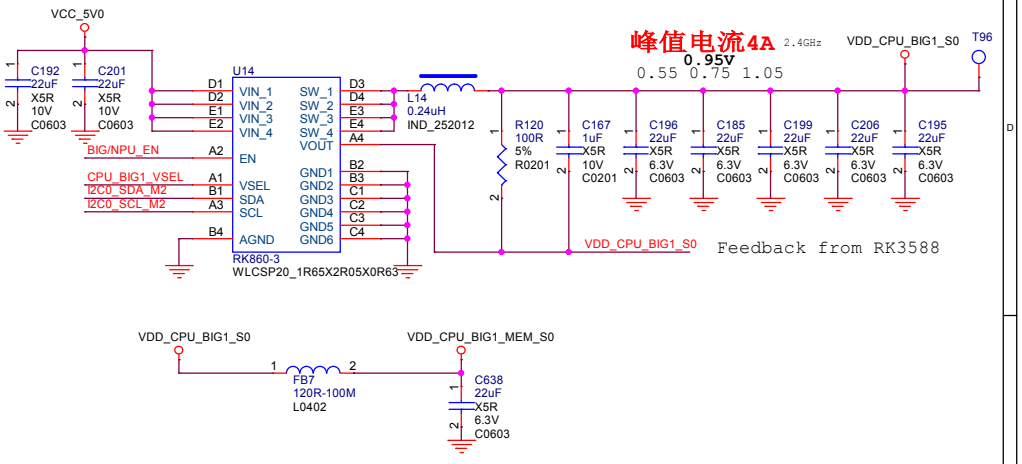


Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 MAX				
File:	11.RTC/FAN				
Date:	Friday, July 25, 2025			Rev:	V1.1
Size:	A3	Sheet:	11	Of:	29

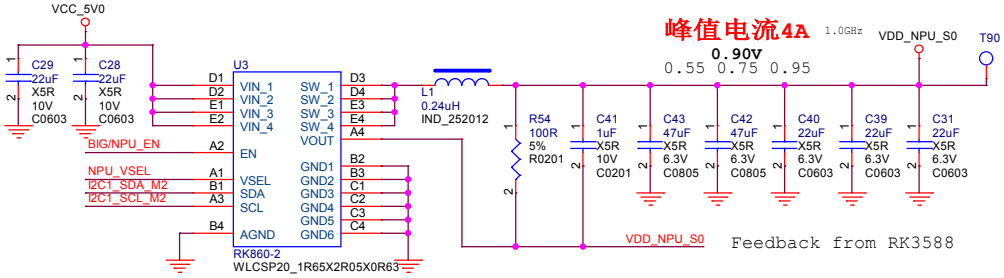
VDD_CPU_BIG0

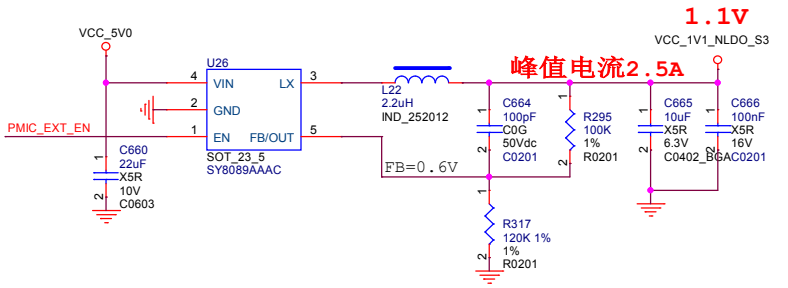


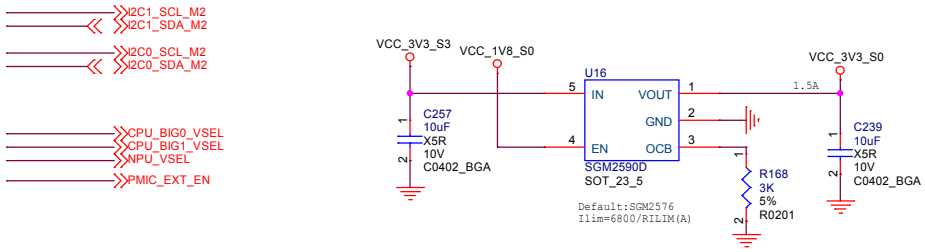
VDD_CPU_BIG1



VDD_NPU

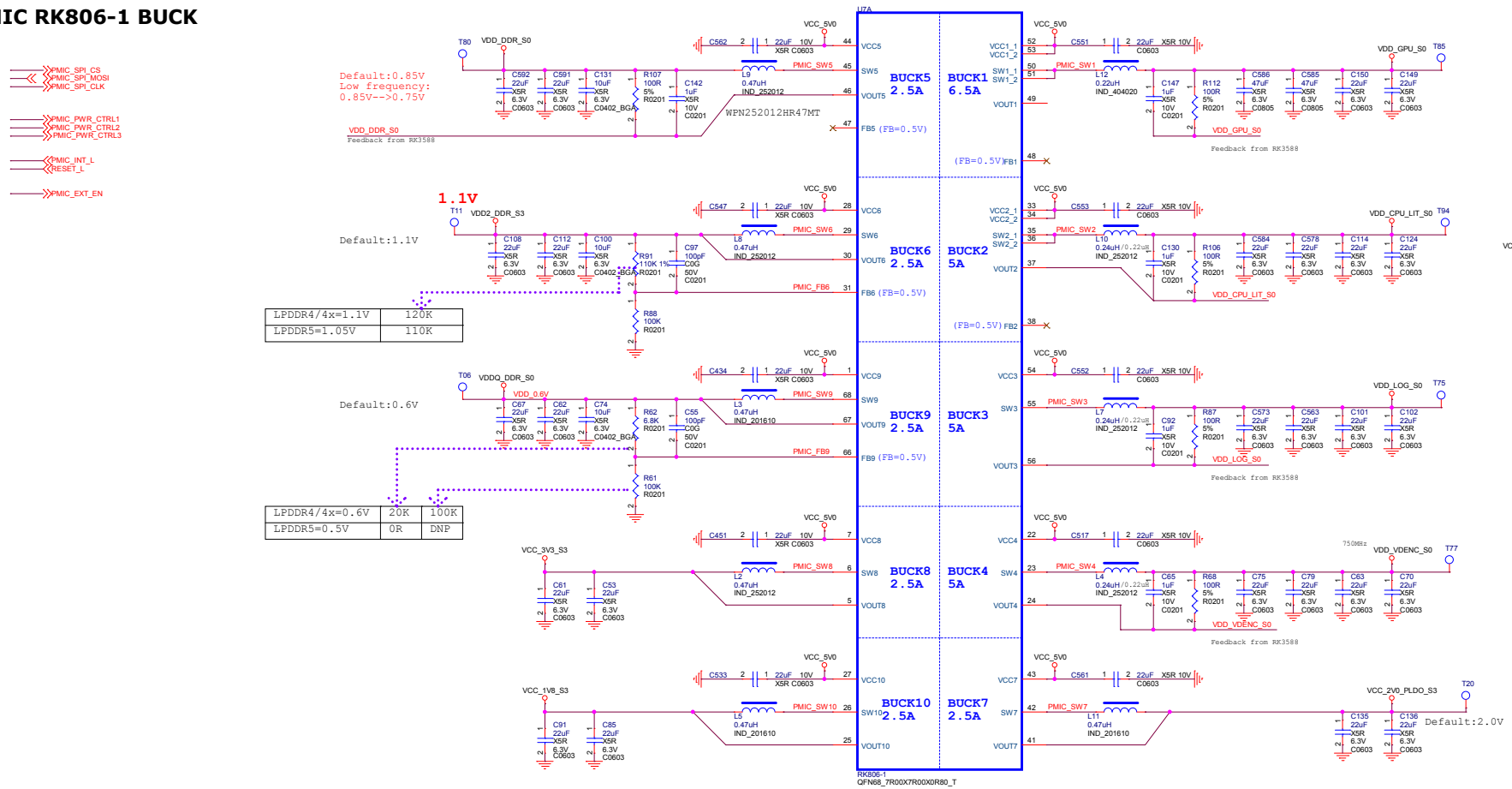




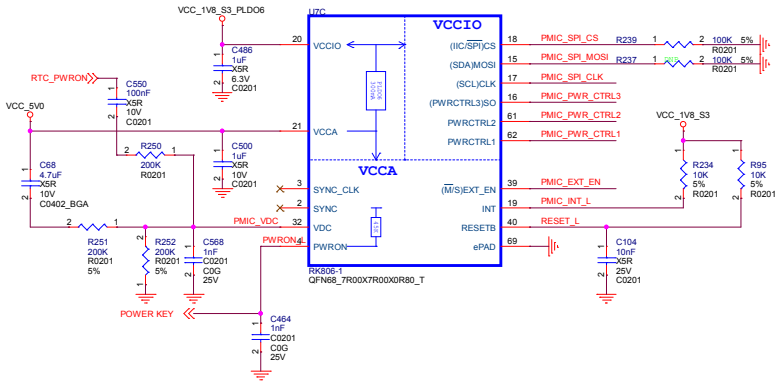


Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 MAX				
File:	12.Power_Ext Discrete				
Date:	Friday, July 25, 2025			Rev:	V1.1
Size:	A3	Sheet:	12	Of:	29

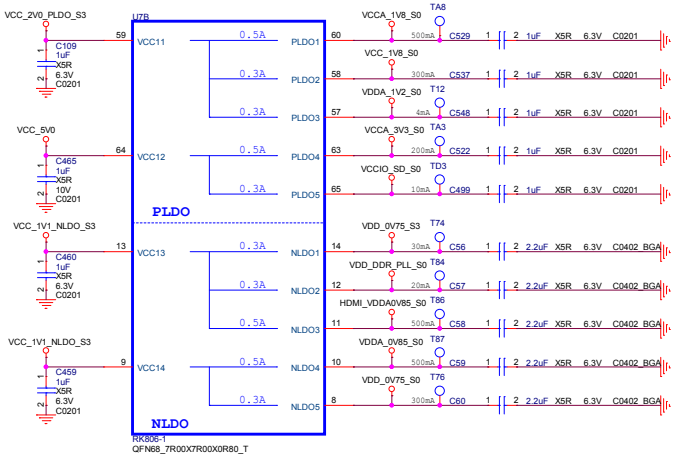
PMIC RK806-1 BUCK



PMIC RK806-1 Managerment

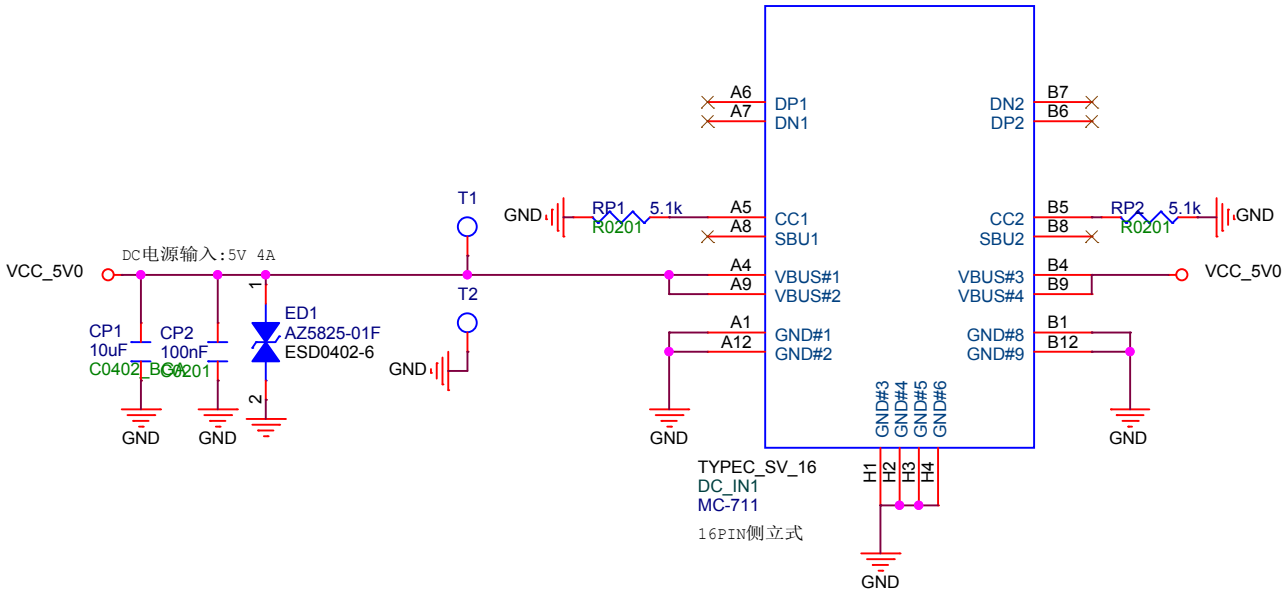


PMIC RK806-1 LDO

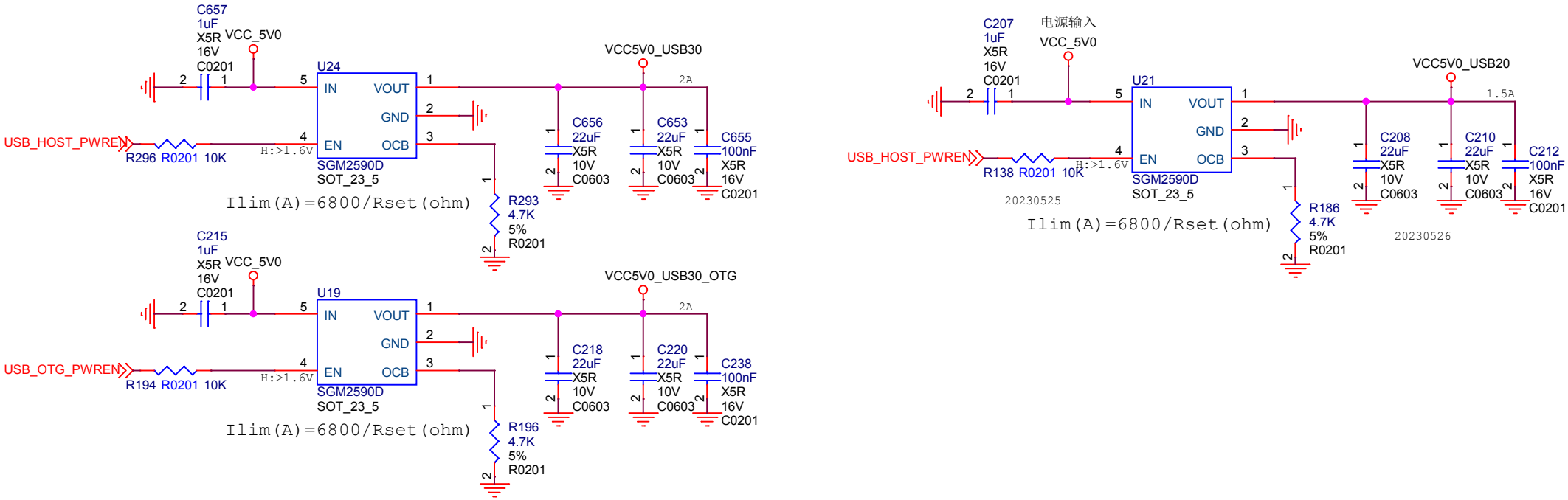


Shenzhen Xunlong Software Co., Ltd	
Project:	OPi 5 MAX
File:	13.Power-PMIC_RK806-1
Date:	Friday, July 25, 2025 Rev: V1.1
Size:	A3 Sheet: 13 Of: 29

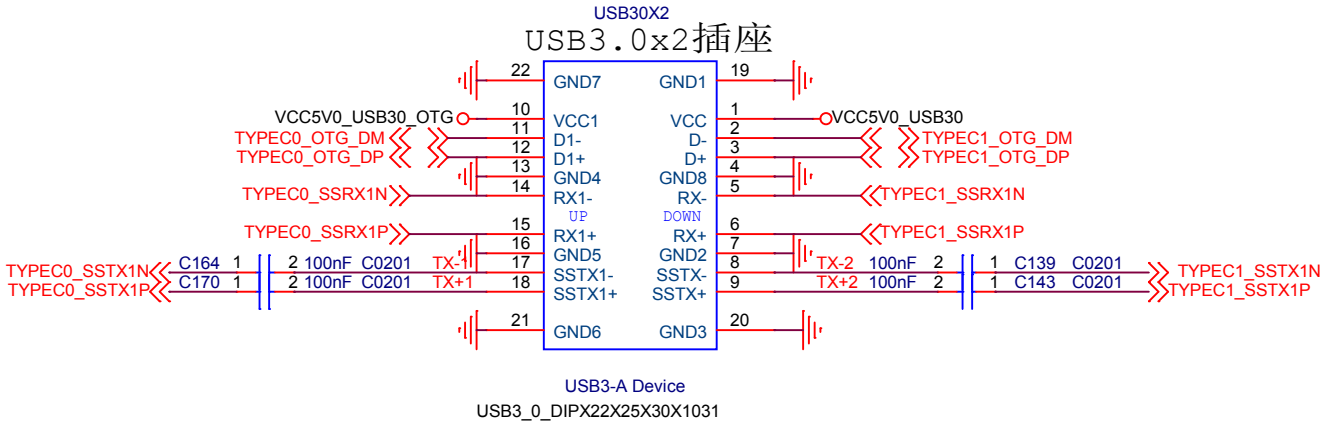
TPYE-C POWER IN

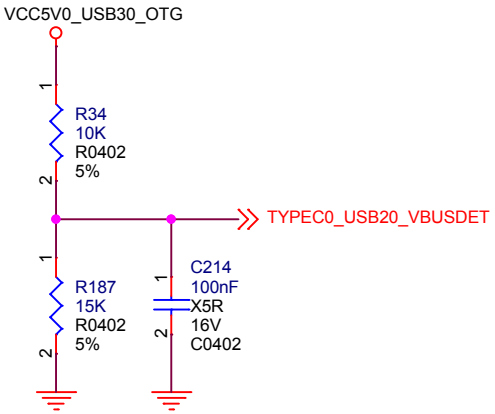


Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 MAX				
File:	14.Type C/DCIN				
Date:	Friday, July 25, 2025			Rev:	V1.1
Size:	A3	Sheet:	14	Of:	29

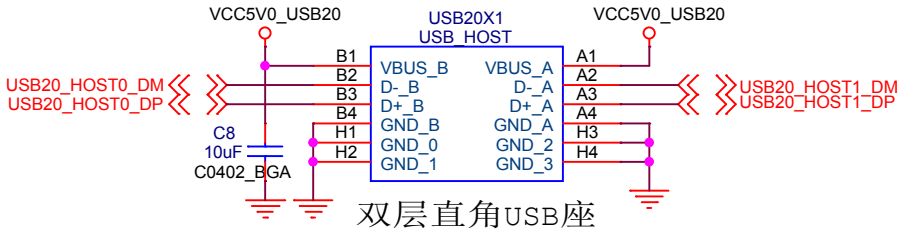


USB3.0*2



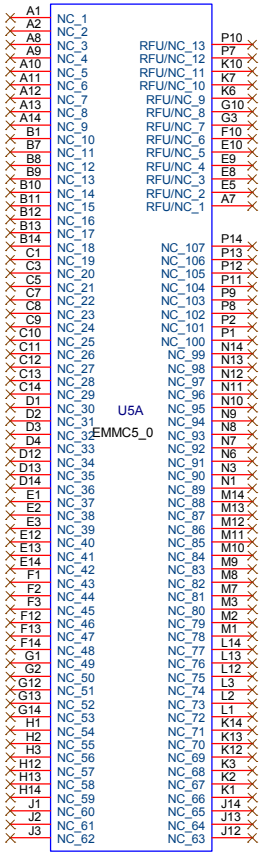


USB2.0*2

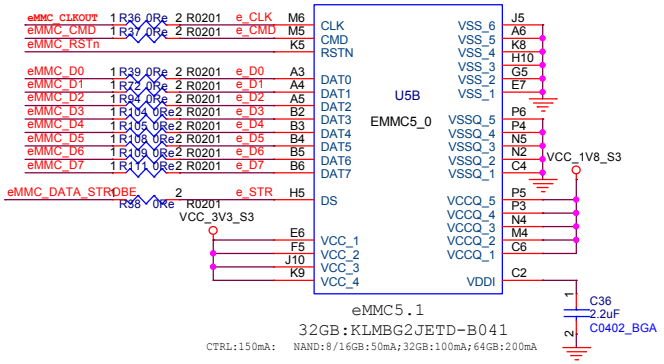


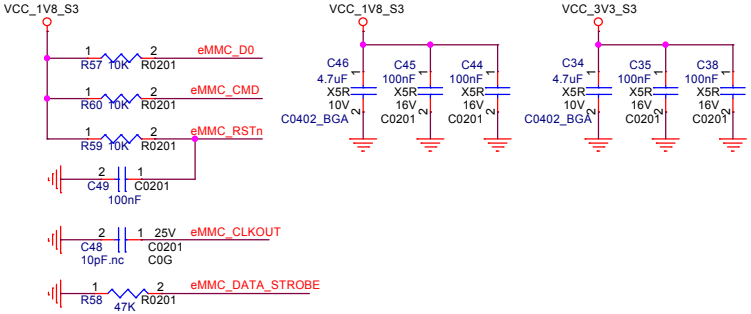
Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 MAX				
File:	15.USB30/USB20				
Date:	Friday, July 25, 2025			Rev:	V1.1
Size:	A3	Sheet:	15	Of:	29

eMMC Flash

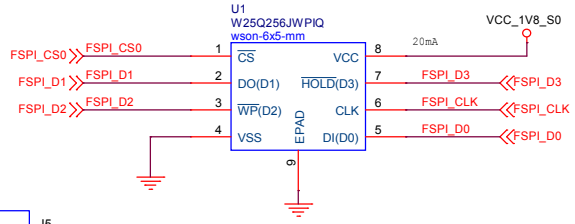


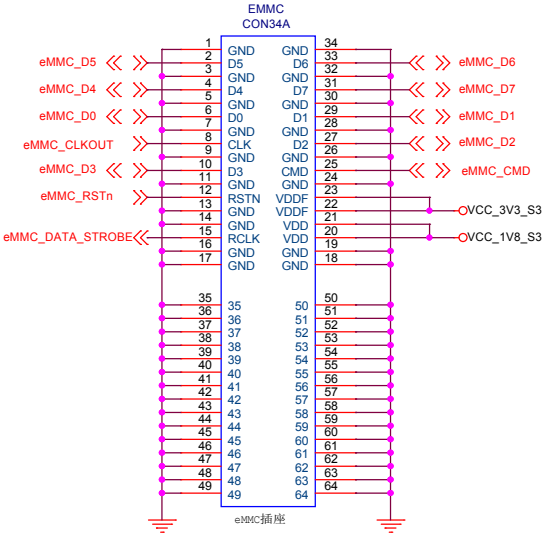






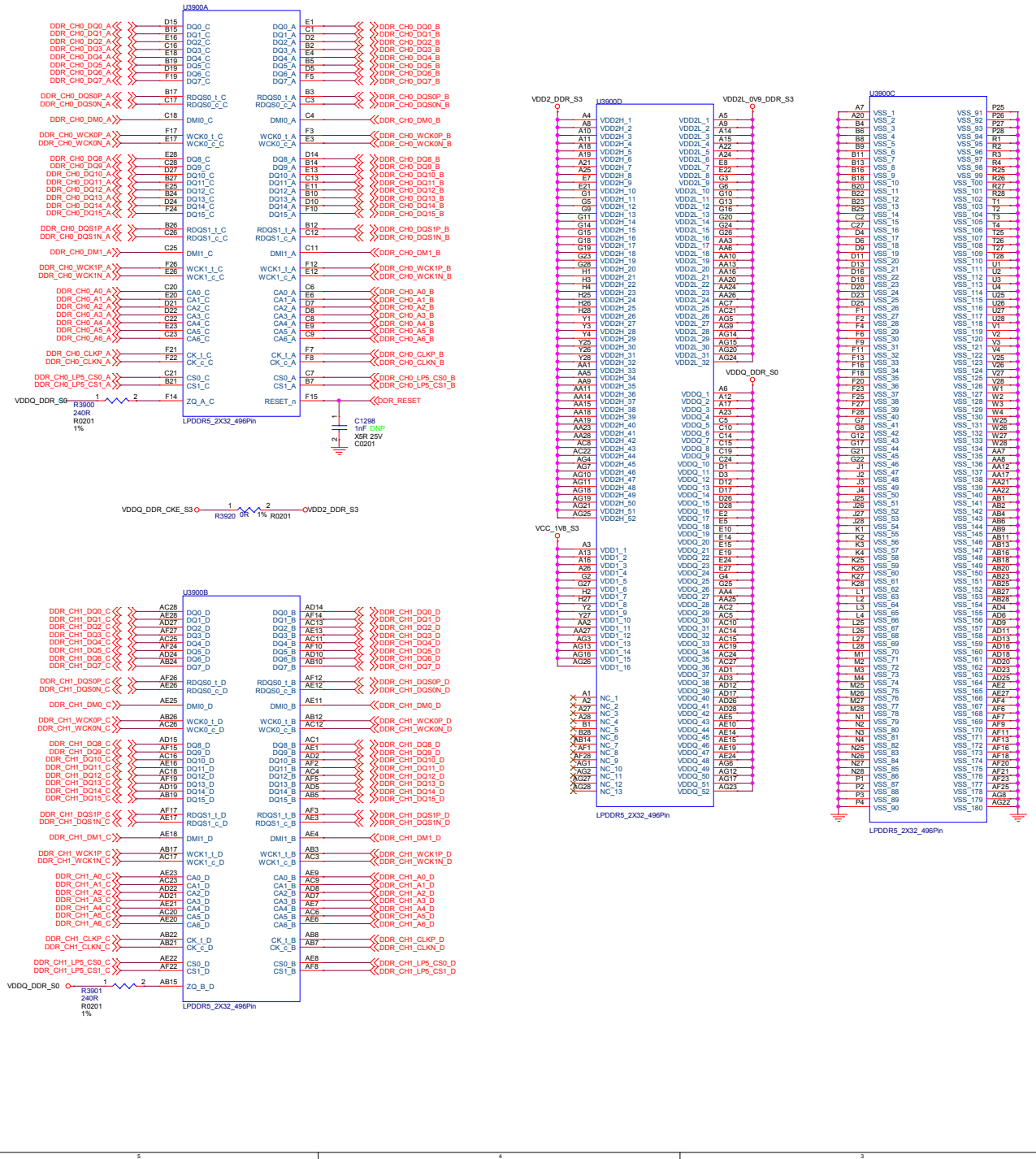
SPI FLASH:1.8V

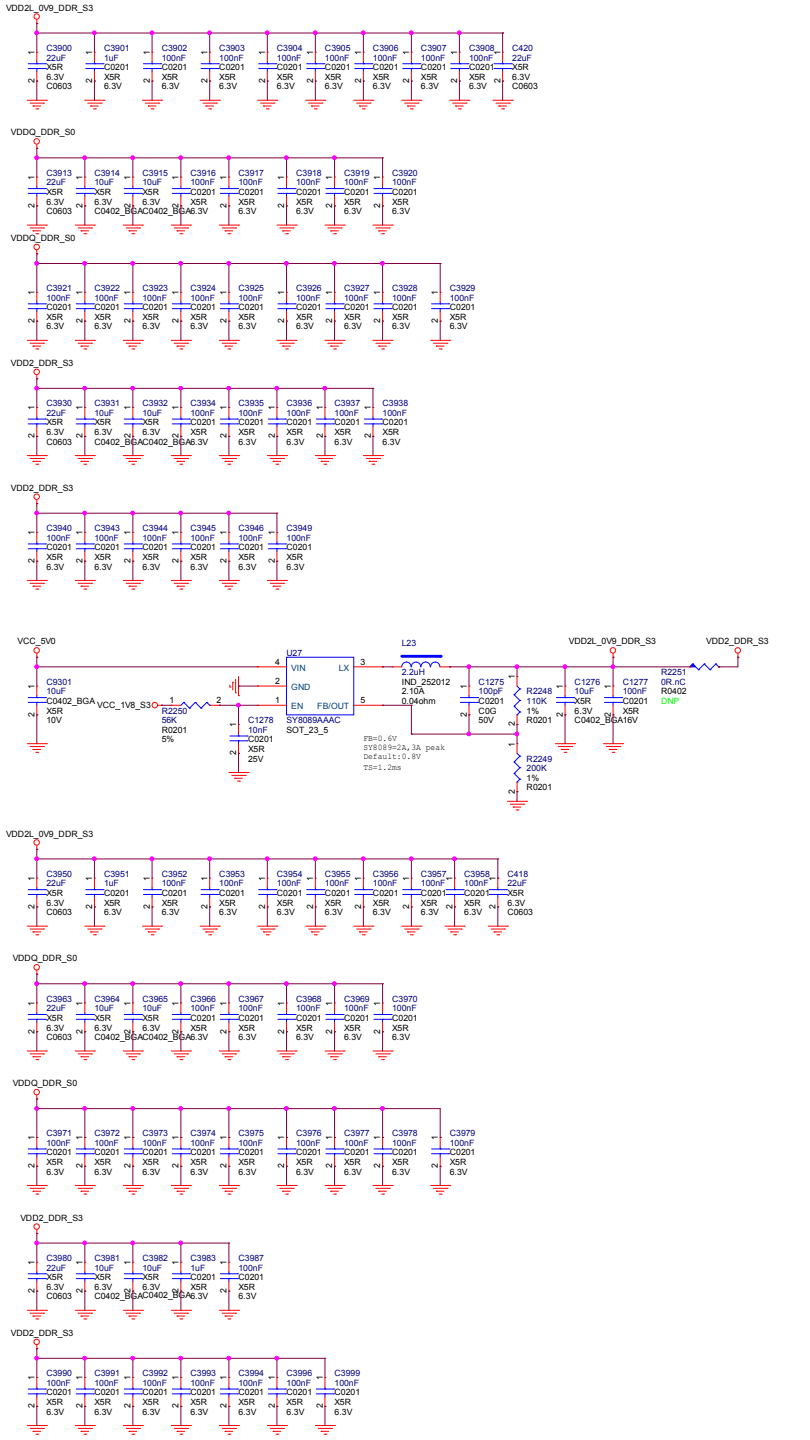




Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 MAX				
File:	16. eMMC/SPI Flash				
Date:	Friday, July 25, 2025			Rev:	V1.1
Size:	A3	Sheet:	16	Of:	29

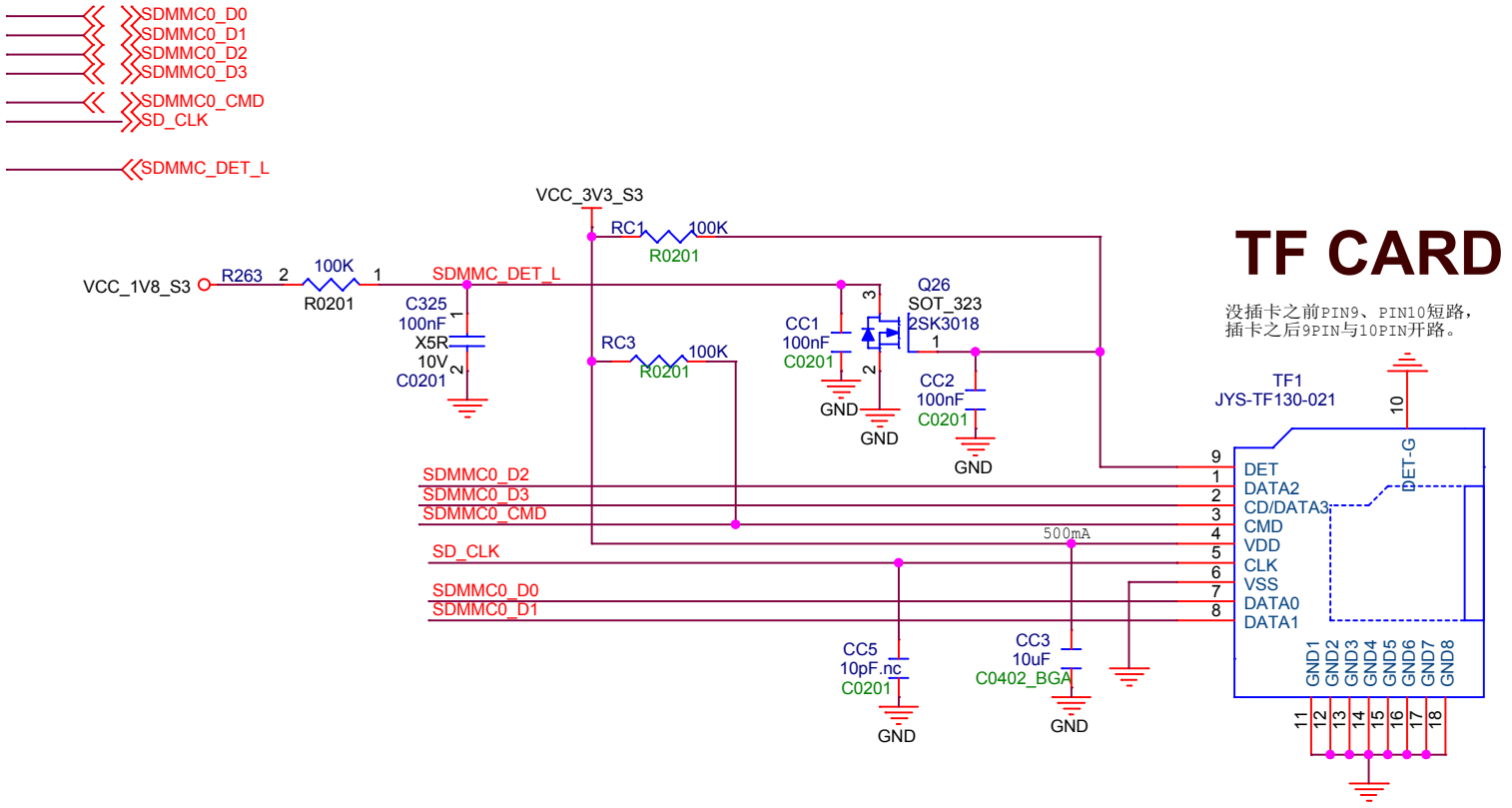
DRAM-LPDDR5X_2X32bit





Shenzhen Xunlong Software Co., Ltd	
Project:	OPi 5 MAX
File:	17.DRAM-LPDDR5X_496P_2X32bit
Date:	Friday, July 25, 2025 Rev: V1.1
Size:	A3 Sheet: 17 Of: 29

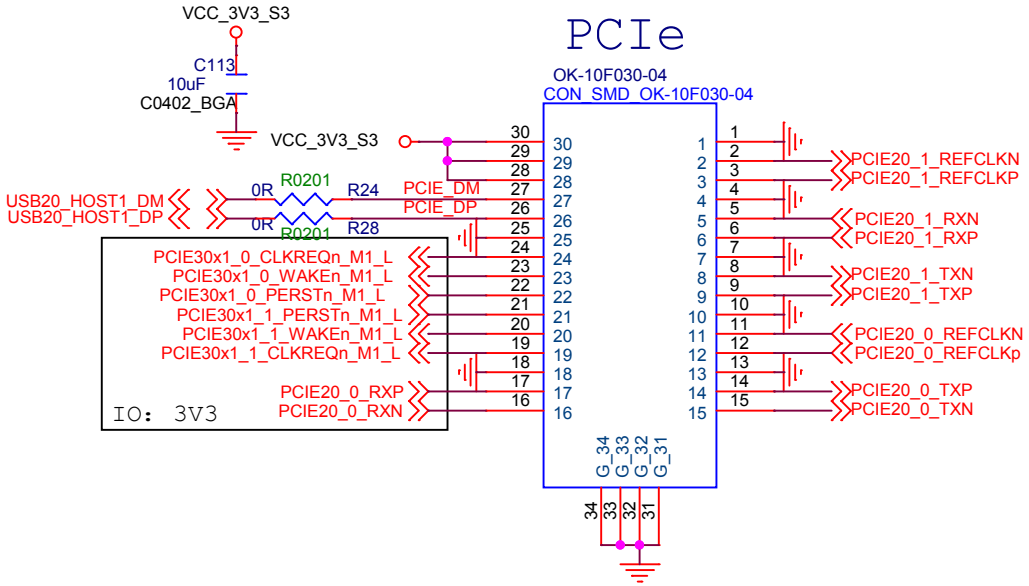
TF CARD



TF CARD

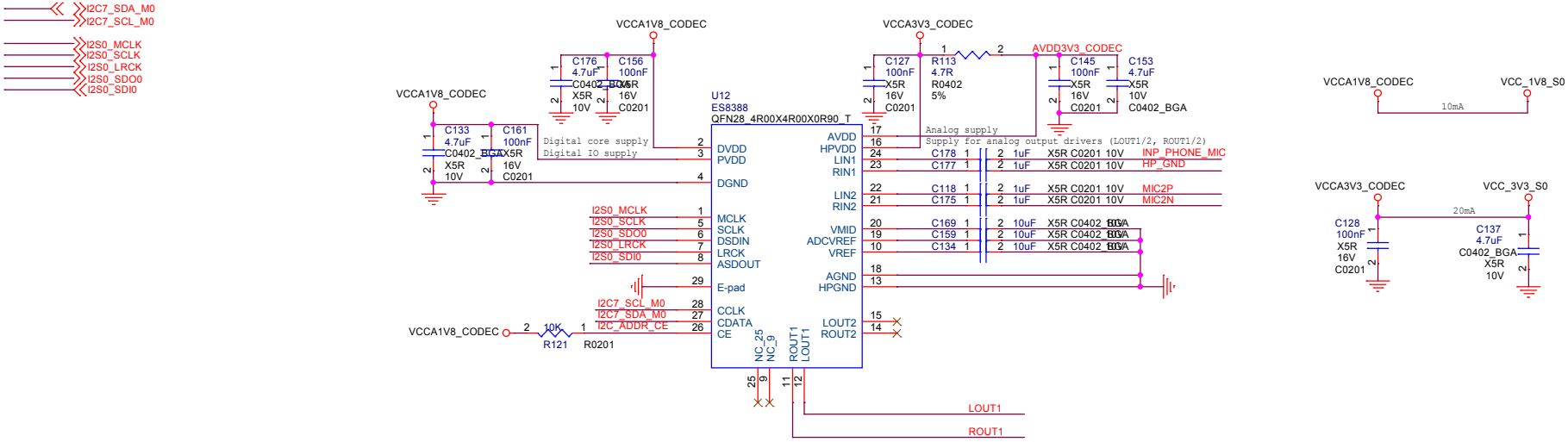
没插卡之前PIN9、PIN10短路，
插卡之后9PIN与10PIN开路。

Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 MAX				
File:	18.Flash-TF Card				
Date:	Friday, July 25, 2025			Rev:	V1.1
Size:	A3	Sheet:	18	Of:	29

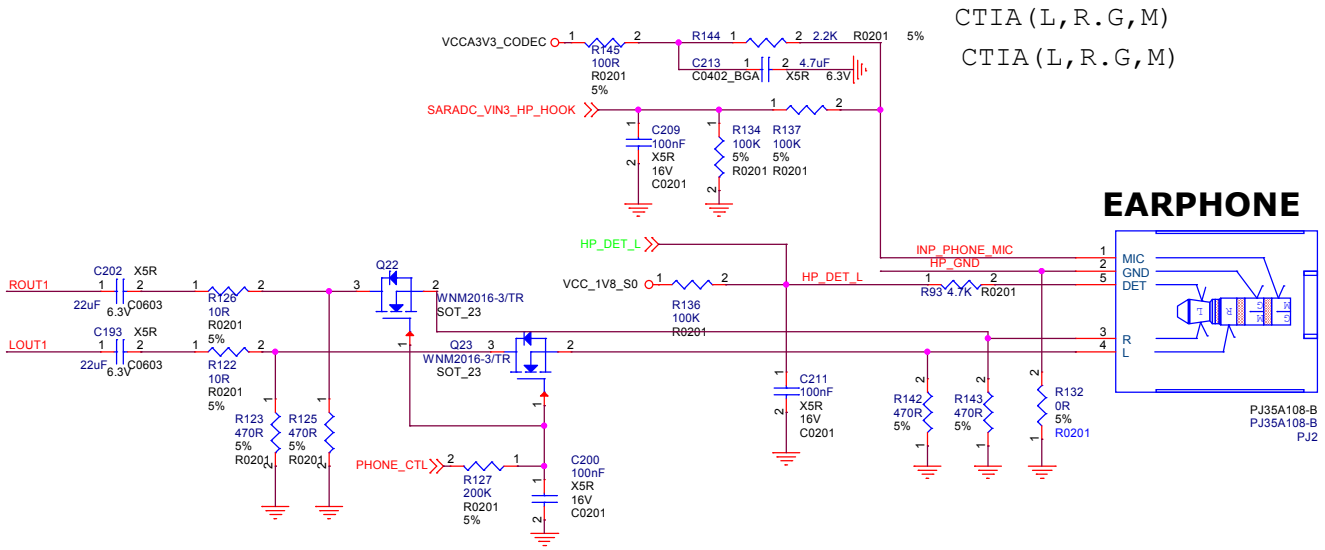


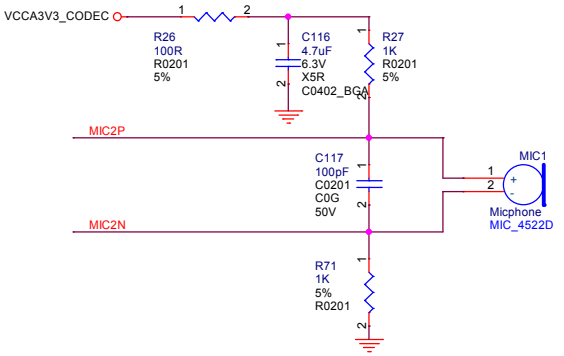
Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 MAX				
File:	19..EXT_30PIN_PCIe				
Date:	Friday, July 25, 2025			Rev:	V1.1
Size:	A3	Sheet:	19	Of:	29

CODEC ES8388



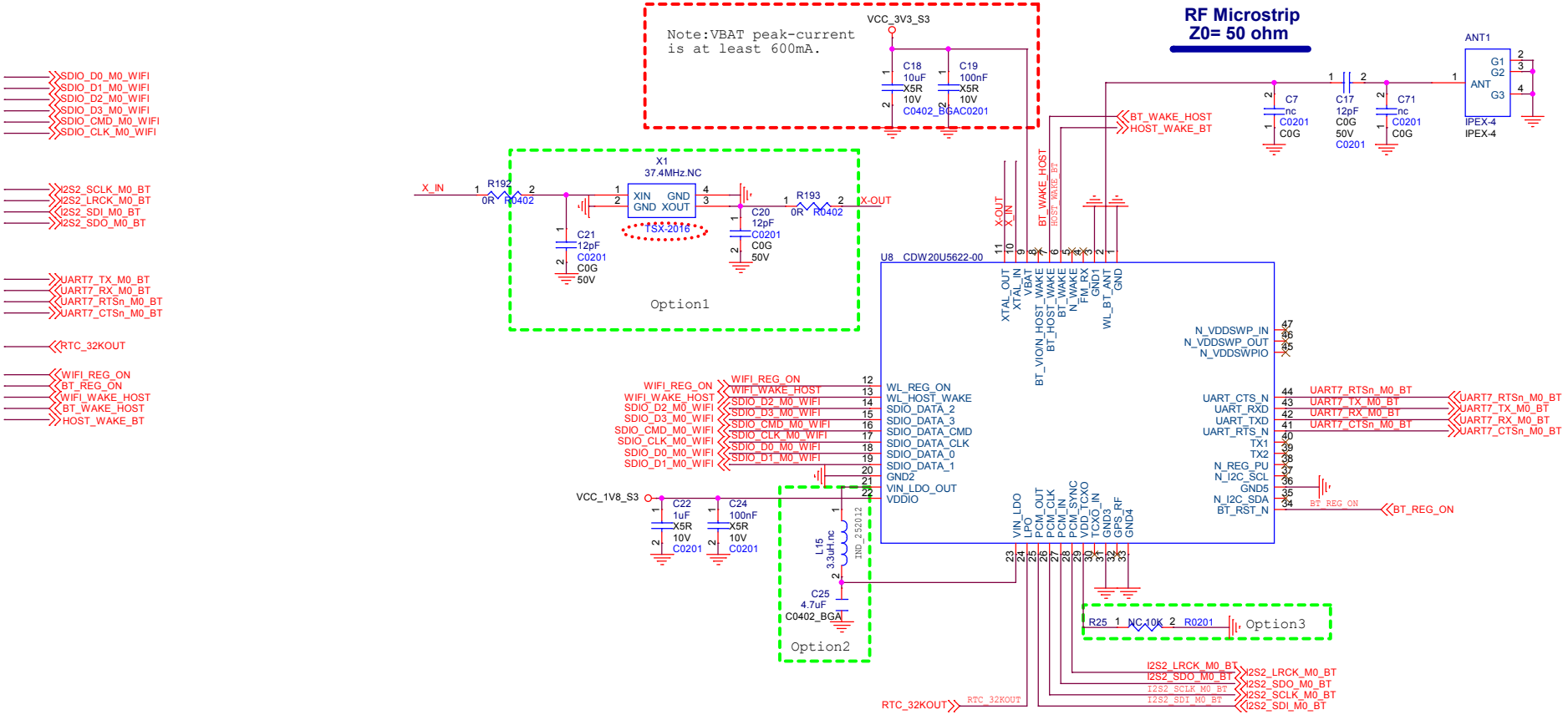
EARPHONE



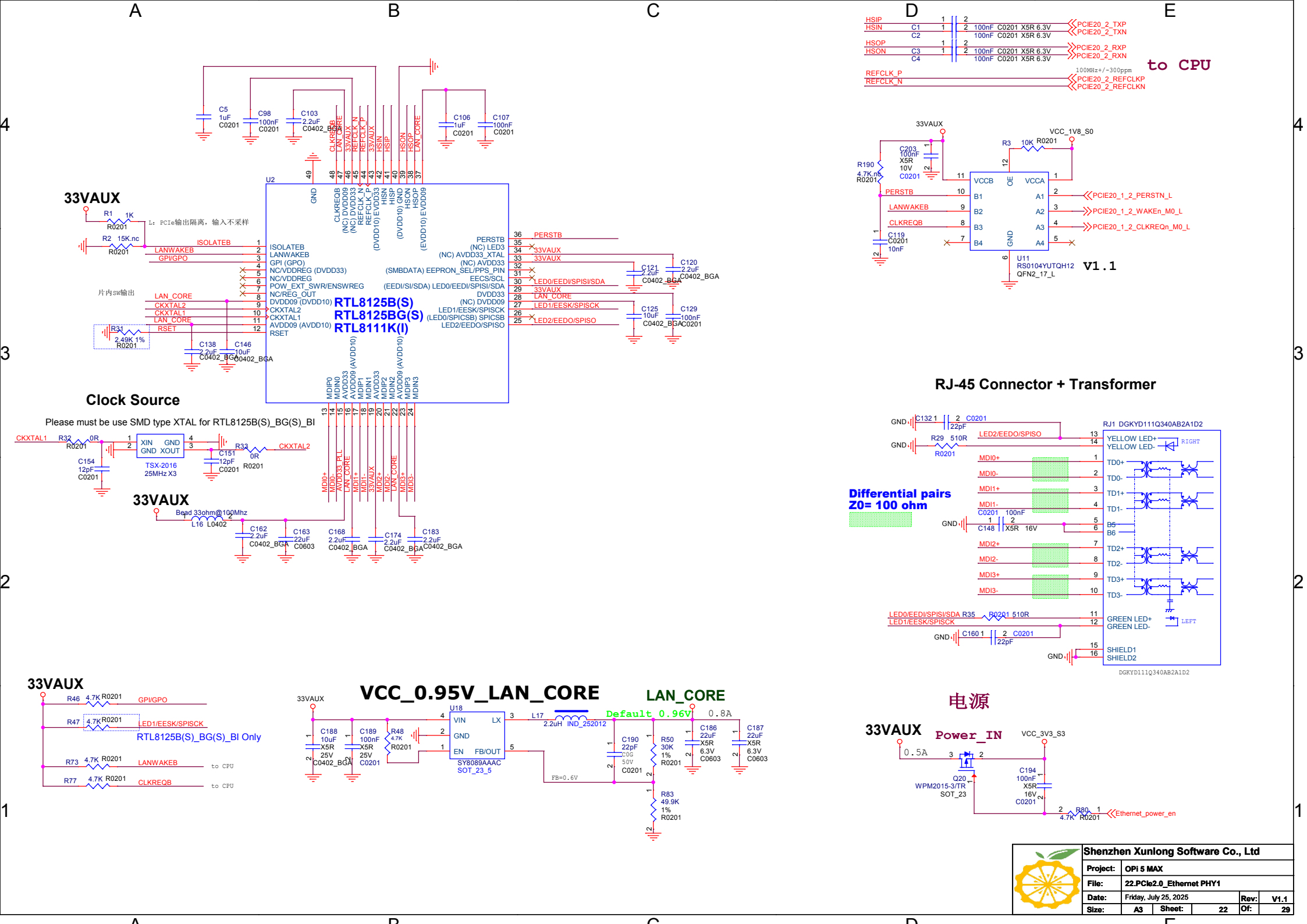


Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 MAX				
File:	20.Audio Codec				
Date:	Friday, July 25, 2025			Rev:	V1.1
Size:	A3	Sheet:	20	Of:	29

SDIO WIFI/BT Module-1T1R

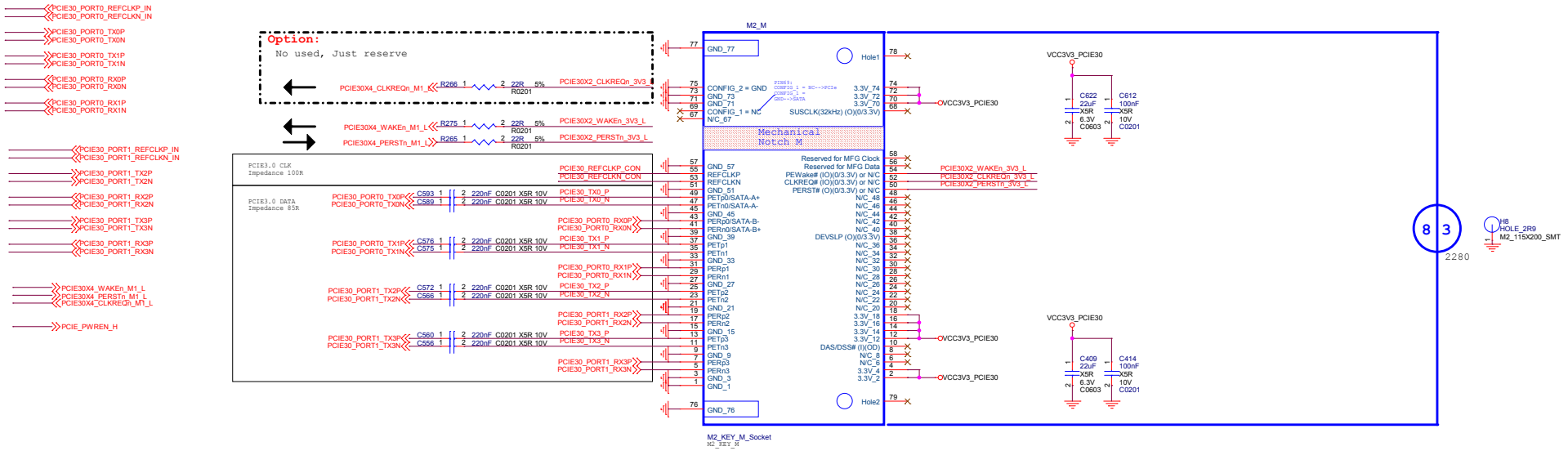


Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 MAX				
File:	21.WIFI/BT-SDIO_1T1R+UART				
Date:	Friday, July 25, 2025			Rev:	V1.1
Size:	A3	Sheet:	21	Of:	29

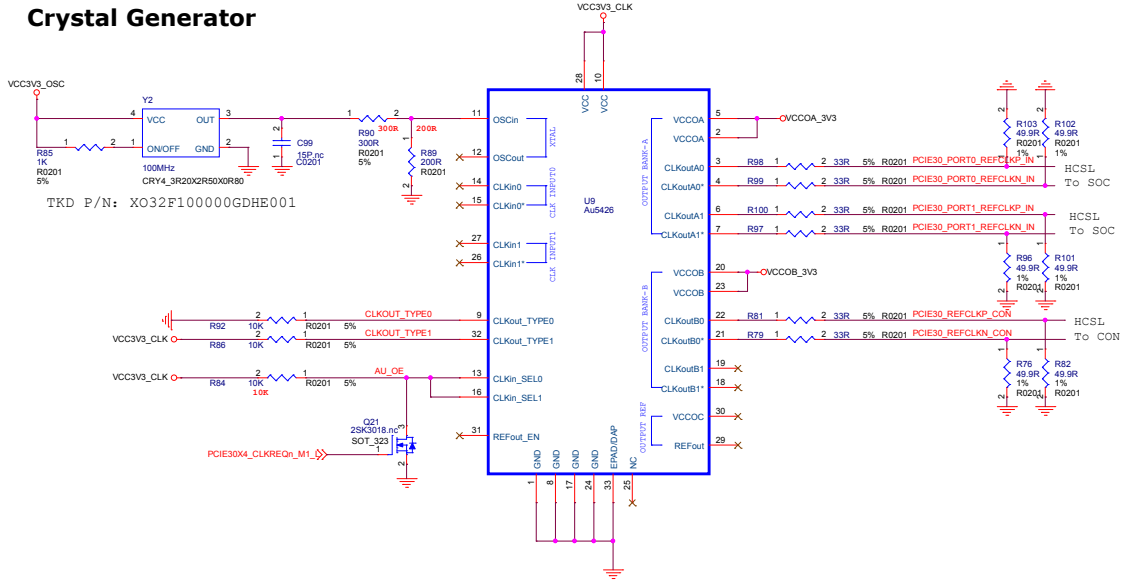


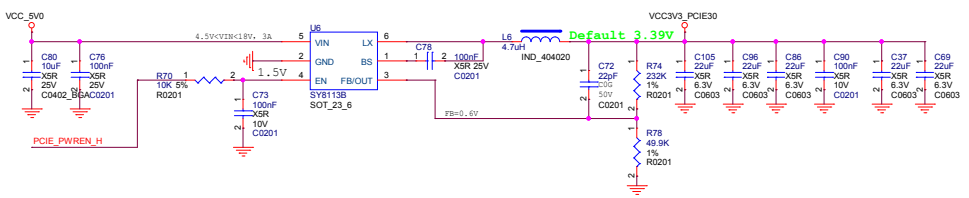
Shenzhen Xunlong Software Co., Ltd	
Project:	OPi 5 MAX
File:	22.PCIe2.0_Ethernet PHY1
Date:	Friday, July 25, 2025
Rev:	V1.1
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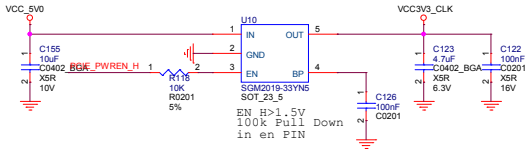
M.2_PCIe3.0 x 2Lanes

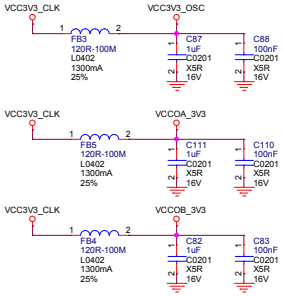


Crystal Generator



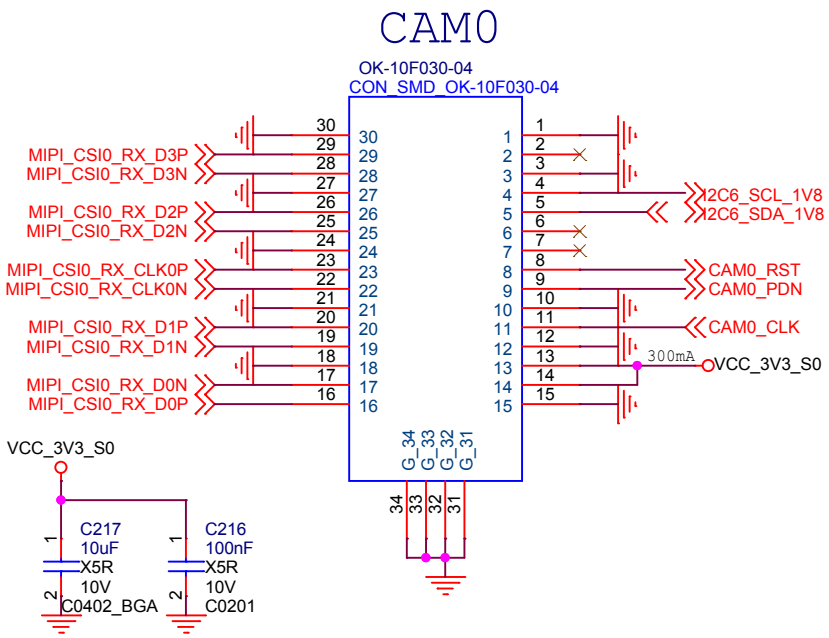


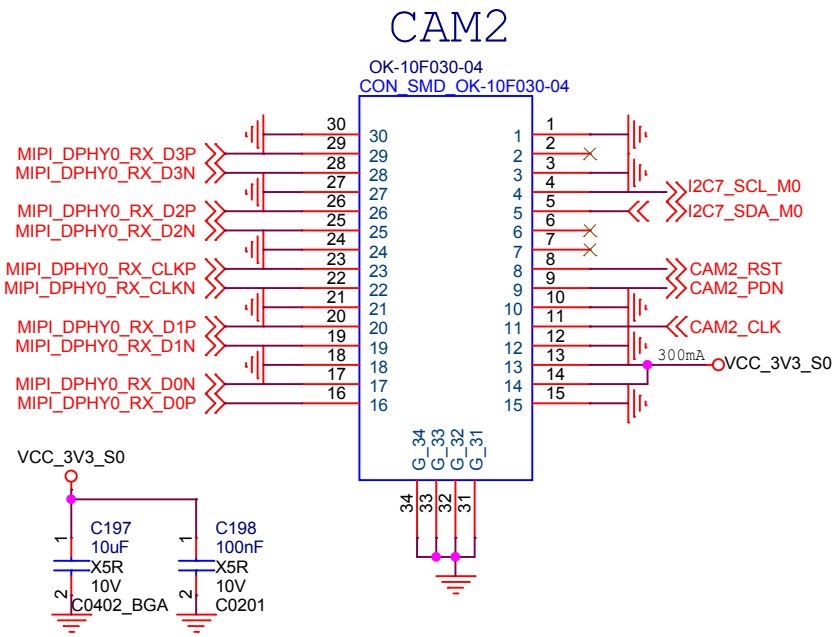


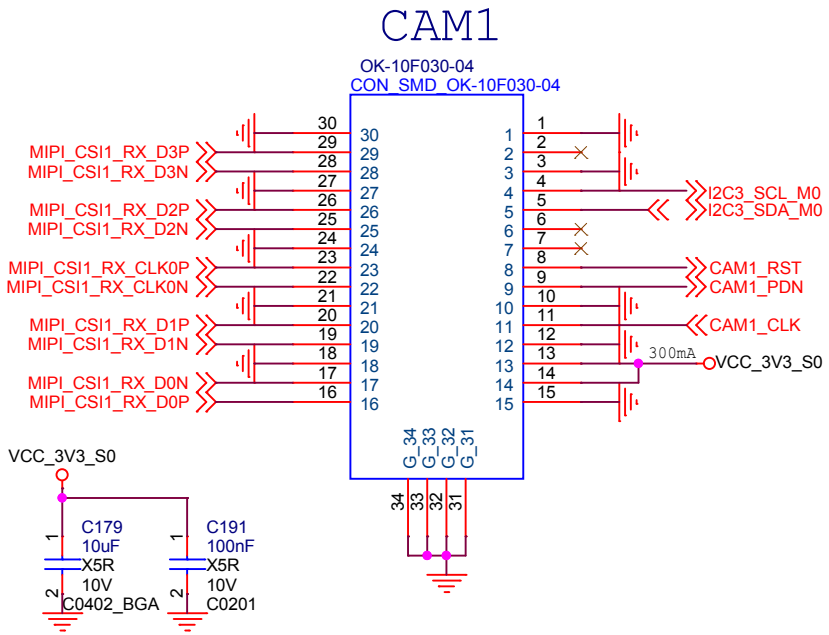


Shenzhen Xunlong Software Co., Ltd			
Project:	OPi 5 MAX		
File:	23.PCIe-M.2 _MKEY _PCIe3.0*4		
Date:	Friday, July 25, 2025	Rev:	V1.1
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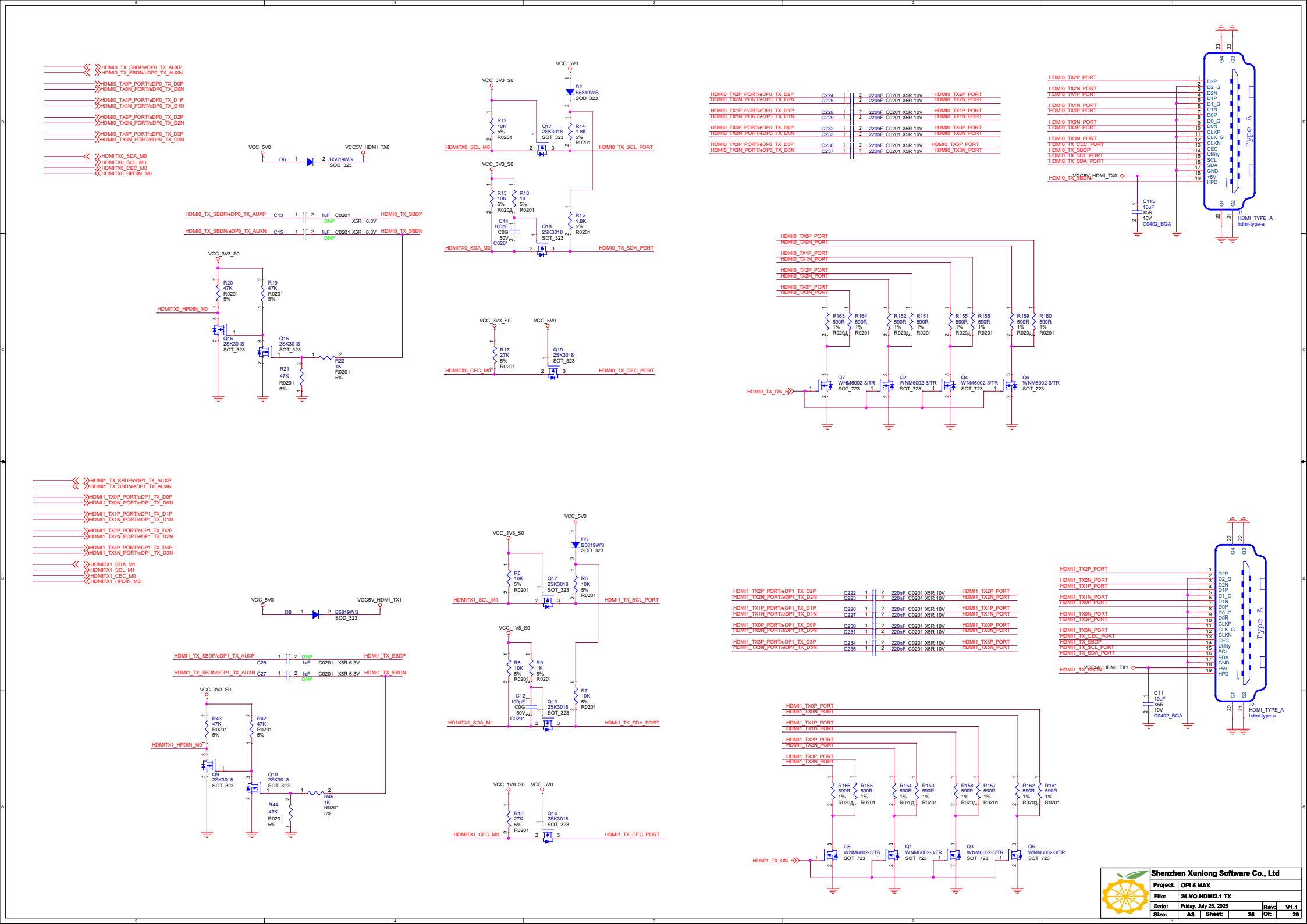
MIPI-CSI0_RX



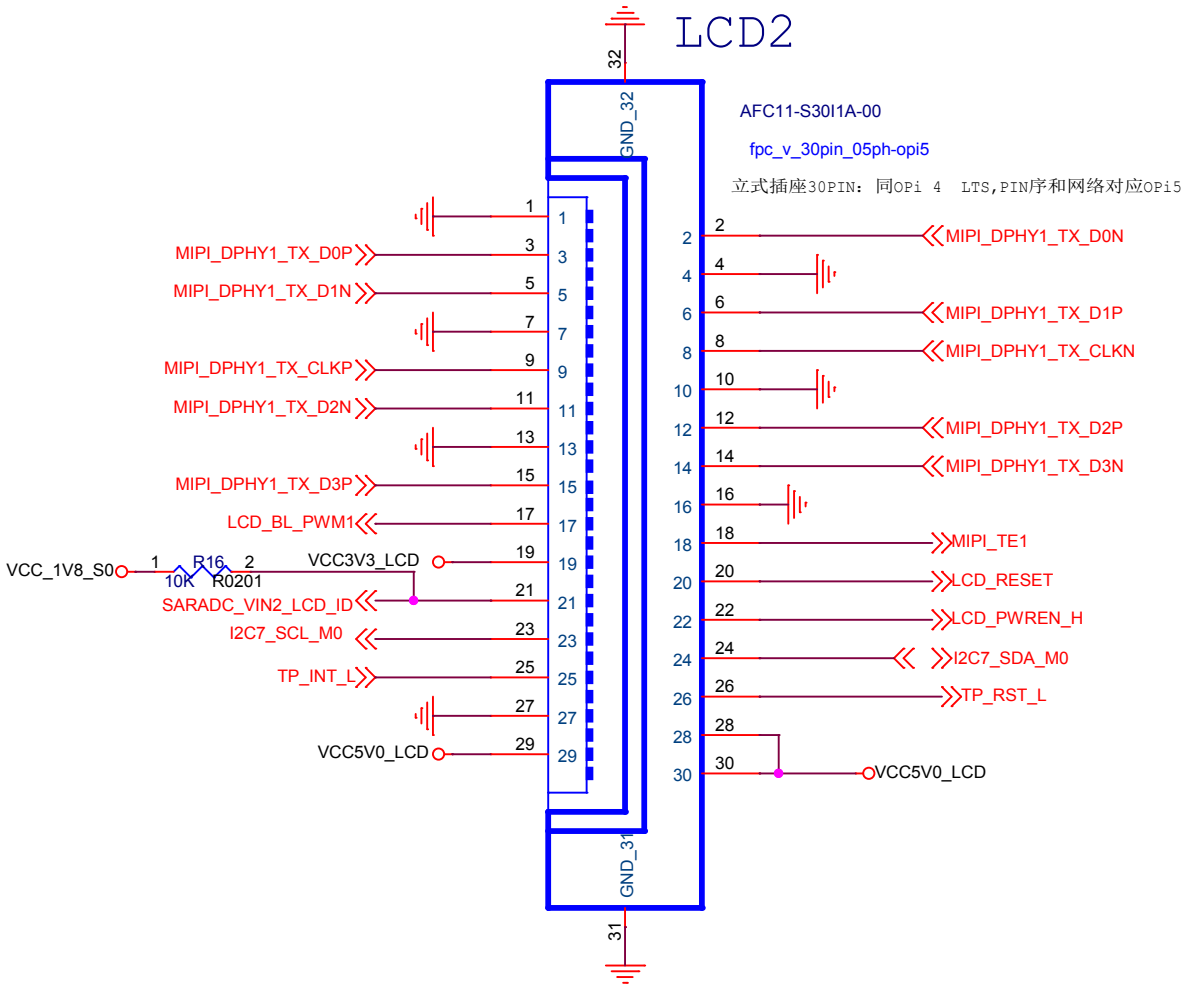




Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 MAX				
File:	24.VI-Camera_MIPI_CSI-RX				
Date:	Friday, July 25, 2025			Rev:	V1.1
Size:	A3	Sheet:	24	Of:	29

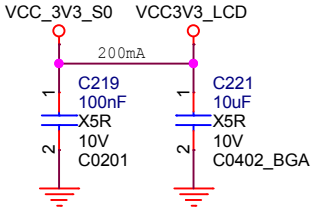

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HDMI0_TX_SBDN/eDP0_TX_AUXN
HDMI0_TX0P_PORT/eDP0_TX_D0P
HDMI0_TX0N_PORT/eDP0_TX_D0N
HDMI0_TX1P_PORT/eDP0_TX_D1P
HDMI0_TX1N_PORT/eDP0_TX_D1N
HDMI0_TX2P_PORT/eDP0_TX_D2P
HDMI0_TX2N_PORT/eDP0_TX_D2N
HDMI0_TX3P_PORT/eDP0_TX_D3P
HDMI0_TX3N_PORT/eDP0_TX_D3N
HDMITX0_SDA_M0
HDMITX0_SCL_M0
HDMITX0_CEC_M0
HDMITX0_HPDIN_M0
VCC_5V0
D6
B5819WS
SOD_323
VCC5V_HDMI_TX0
HDMI0_TX_SBDP/eDP0_TX_AUXP
C13
1uF
C0201
DNP
X5R
6.3V
HDMI0_TX_SBDP
HDMI0_TX_SBDN/eDP0_TX_AUXN
C15
HDMI0_TX_SBDN
VCC_3V3_S0
R20
47K
R0201
5%
R19
Q16
2SK3018
SOT_323
Q15
R21
R22
1K
HDMITX0_SCL_M0
R12
10K
Q17
R14
1.8K
HDMI0_TX_SCL_PORT
R13
R18
C14
100pF
C0G
50V
Q18
R15
HDMI0_TX_SDA_PORT
R17
27K
Q19
HDMI0_TX_CEC_PORT
HDMI0_TX2P_PORT/eDP0_TX_D2P
HDMI0_TX2N_PORT/eDP0_TX_D2N
HDMI0_TX1P_PORT/eDP0_TX_D1P
HDMI0_TX1N_PORT/eDP0_TX_D1N
HDMI0_TX0P_PORT/eDP0_TX_D0P
HDMI0_TX0N_PORT/eDP0_TX_D0N
HDMI0_TX3P_PORT/eDP0_TX_D3P
HDMI0_TX3N_PORT/eDP0_TX_D3N
C224
C225
C228
C229
C232
C233
C236
C237
220nF
C0201
X5R
10V
HDMI0_TX2P_PORT
HDMI0_TX2N_PORT
HDMI0_TX1P_PORT
HDMI0_TX1N_PORT
HDMI0_TX0P_PORT
HDMI0_TX0N_PORT
HDMI0_TX3P_PORT
HDMI0_TX3N_PORT
HDMI0_TX_CEC_PORT
HDMI0_TX_SBDP
HDMI0_TX_SCL_PORT
HDMI0_TX_SDA_PORT
HDMI0_TX_SBDN
VCC5V_HDMI_TX0
C115
10uF
X5R
10V
C0402_BGA
J1
HDMI_TYPE_A
hdmi-type-a
HDMI0_TX0P_PORT
HDMI0_TX0N_PORT
HDMI0_TX1P_PORT
HDMI0_TX1N_PORT
HDMI0_TX2P_PORT
HDMI0_TX2N_PORT
HDMI0_TX3P_PORT
HDMI0_TX3N_PORT
R163
590R
1%
R0201
R164
R152
R151
R155
R156
R159
R160
HDMI0_TX_ON_H
Q7
WNM6002-3/TR
SOT_723
Q2
Q4
Q6
HDMI1_TX_SBDP/eDP1_TX_AUXP
HDMI1_TX_SBDN/eDP1_TX_AUXN
HDMI1_TX0P_PORT/eDP1_TX_D0P
HDMI1_TX0N_PORT/eDP1_TX_D0N
HDMI1_TX1P_PORT/eDP1_TX_D1P
HDMI1_TX1N_PORT/eDP1_TX_D1N
HDMI1_TX2P_PORT/eDP1_TX_D2P
HDMI1_TX2N_PORT/eDP1_TX_D2N
HDMI1_TX3P_PORT/eDP1_TX_D3P
HDMI1_TX3N_PORT/eDP1_TX_D3N
HDMITX1_SDA_M1
HDMITX1_SCL_M1
HDMITX1_CEC_M0
HDMITX1_HPDIN_M0
VCC_5V0
D8
B5819WS
SOD_323
VCC5V_HDMI_TX1
HDMI1_TX_SBDP/eDP1_TX_AUXP
C26
DNP
1uF
C0201
X5R
6.3V
HDMI1_TX_SBDP
HDMI1_TX_SBDN/eDP1_TX_AUXN
C27
HDMI1_TX_SBDN
VCC_3V3_S0
R43
47K
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5%
R42
HDMITX1_HPDIN_M0
Q9
2SK3018
SOT_323
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R44
R45
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VCC_1V8_S0
D5
B5819WS
SOD_323
R5
10K
R6
HDMITX1_SCL_M1
Q12
HDMI1_TX_SCL_PORT
R8
R9
1K
C12
100pF
C0G
50V
C0201
Q13
R7
HDMITX1_SDA_M1
HDMI1_TX_SDA_PORT
R10
27K
Q14
HDMITX1_CEC_M0
HDMI1_TX_CEC_PORT
HDMI1_TX2P_PORT/eDP1_TX_D2P
HDMI1_TX2N_PORT/eDP1_TX_D2N
HDMI1_TX1P_PORT/eDP1_TX_D1P
HDMI1_TX1N_PORT/eDP1_TX_D1N
HDMI1_TX0P_PORT/eDP1_TX_D0P
HDMI1_TX0N_PORT/eDP1_TX_D0N
HDMI1_TX3P_PORT/eDP1_TX_D3P
HDMI1_TX3N_PORT/eDP1_TX_D3N
C222
C223
C226
C227
C230
C231
C234
C235
220nF
C0201
X5R
10V
HDMI1_TX2P_PORT
HDMI1_TX2N_PORT
HDMI1_TX1P_PORT
HDMI1_TX1N_PORT
HDMI1_TX0P_PORT
HDMI1_TX0N_PORT
HDMI1_TX3P_PORT
HDMI1_TX3N_PORT
HDMI1_TX_CEC_PORT
HDMI1_TX_SBDP
HDMI1_TX_SCL_PORT
HDMI1_TX_SDA_PORT
HDMI1_TX_SBDN
VCC5V_HDMI_TX1
C11
10uF
X5R
10V
C0402_BGA
J2
HDMI_TYPE_A
hdmi-type-a
R166
590R
1%
R0201
R165
R154
R153
R158
R157
R162
R161
HDMI1_TX_ON_H
Q8
WNM6002-3/TR
SOT_723
Q1
Q3
Q5
Shenzhen Xunlong Software Co., Ltd
Project: OPi 5 MAX
File: 25.VO-HDMI2.1 TX
Date: Friday, July 25, 2025
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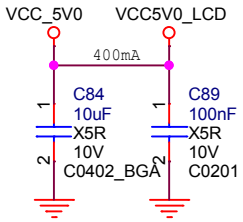
MIPI DPHY1 TX



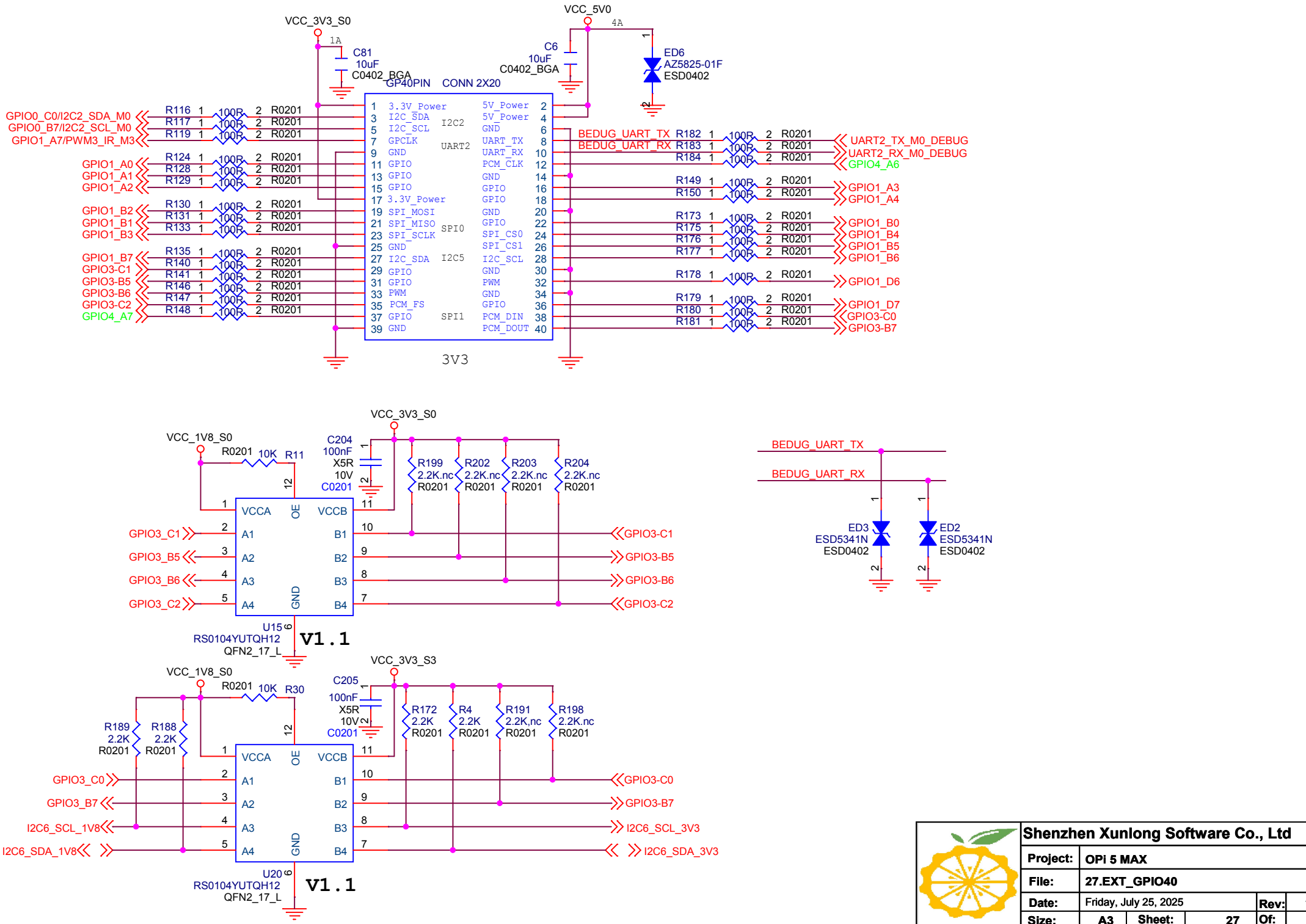
FPC Pin List

Pin1 :GND
Pin2 :D0N
Pin3 :D0P
Pin4 :GND
Pin5 :D1N
Pin6 :D1P
Pin7 :GND
Pin8 :CLKN/AUXN
Pin9 :CLKP/AUXP
Pin10:GND
Pin11:D2N
Pin12:D2P
Pin13:GND
Pin14:D3N
Pin15:D3P
Pin16:GND
Pin17:LCD_PWM_BL
Pin18:LCD_TE
Pin19:VCC3V3_LCD
Pin20:LCD_RST
Pin21:LCD_ID
Pin22:LCD_PWREN
Pin23:TP_I2C_SCL
Pin24:TP_I2C_SDA
Pin25:TP_INT
Pin26:TP_RST
Pin27:GND
Pin28:5V0
Pin29:5V0
Pin30:5V0





Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 MAX				
File:	26.VO-LCM_MIPI				
Date:	Friday, July 25, 2025			Rev:	V1.1
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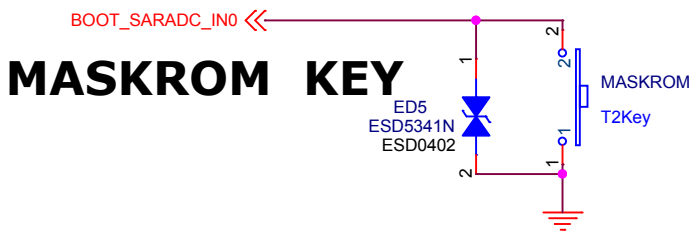


Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 MAX				
File:	27.EXT_GPIO40				
Date:	Friday, July 25, 2025			Rev:	V1.1
Size:	A3	Sheet:	27	Of:	29

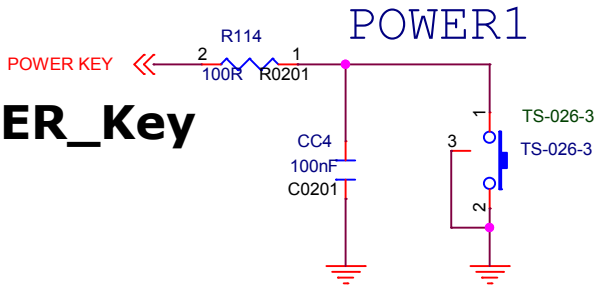
5 4 3 2 1

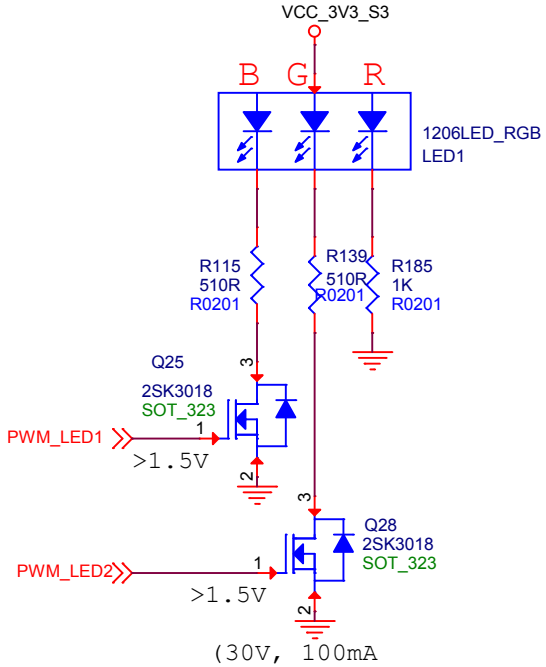
D C B A

BOOT



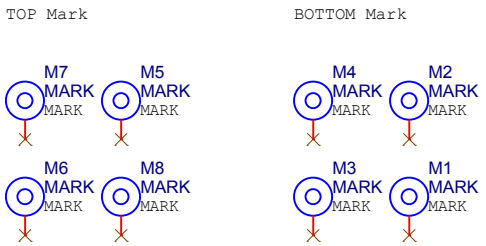
POWER1



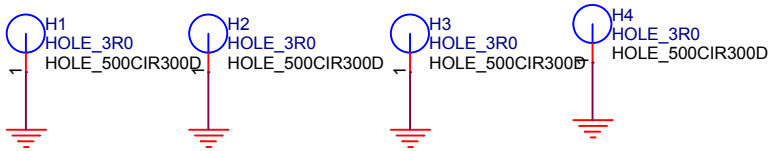


Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 MAX				
File:	28. KEY/LED				
Date:	Friday, July 25, 2025			Rev:	V1.1
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Mark



HOLE



HEATSINK

Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 MAX				
File:	29.Mark/Hole/Heatsink				
Date:	Friday, July 25, 2025			Rev:	V1.1
Size:	A3	Sheet:	29	Of:	29